Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: DESIGN CALCULATION FOR TANK FOUNDATIONS - 49W1 Raw Water Treatment Plant

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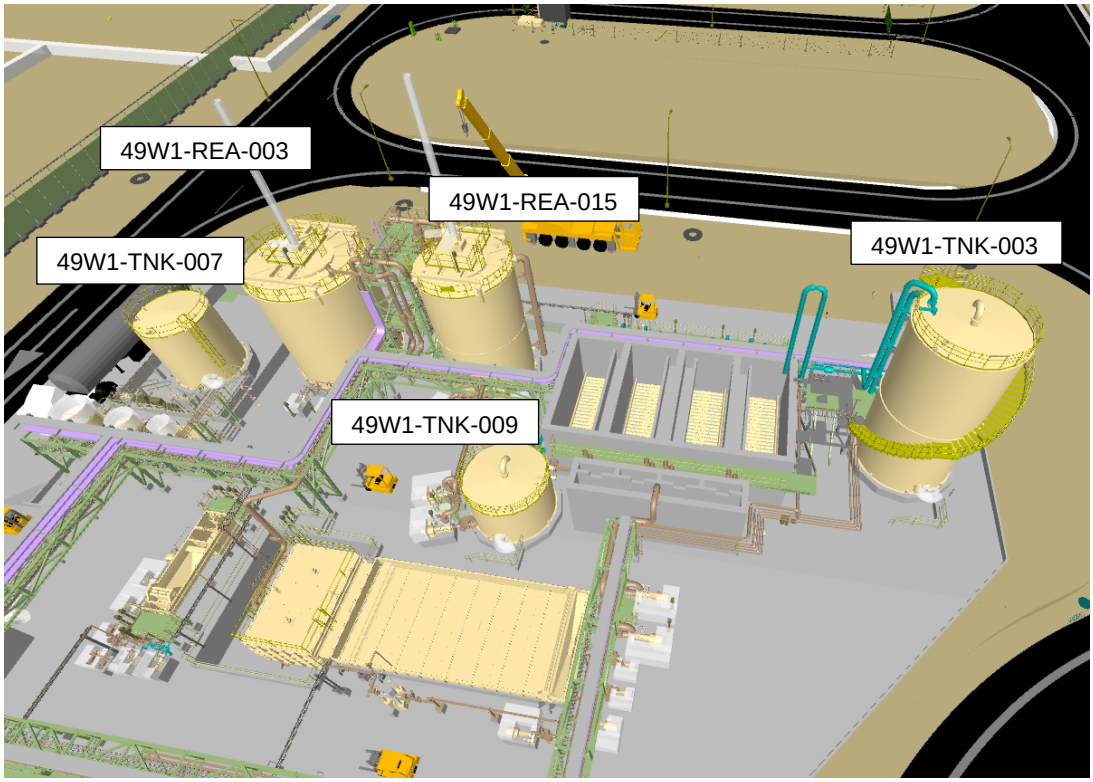
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1. INTENT

This document is to Prepare for design of reactor tank foundations. The analysis & design is carried out in accordance with project design criteria, building codes & standards. The design calculation ensures the safety and serviceability.

2. SCOPE



The design follows and verified with requirements in Saudi Building Code:

- SBC_301 (Structural – Loading and Forces)
- SBC_304 (Structural – Concrete Structures)
- SBC_303 (Structural – Soil and Foundations)

Calculations are carried out with Dlubal RFEM 5.21.

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3. MATERIAL PROPERTY

- concrete: C35, f_{ck} = 35MPa
- reinforcement: ASTM A 615/A 615 M Grade 60, f_{yk}=420MPa
- Anchor bolt: ASTM F1554 G36

4. LOAD CALCULATIONS

4.1. Design load

Basic load cases included in the design, as suggested in SBC 301, are:

1. Self-weight of the tank (D)
2. Liquid load (F) at operation level and max level
3. Wind load (W)
4. Platform load (D)
5. Seismic load (E)

4.1.1. Dead load from tank

The tank weight is distributed to the ring wall.

4.1.2. Liquid load

Liquid load is considered for both cases: operational weight and max weight.

4.1.3. External platform load

Platform load is distributed to ring wall.

4.1.4. Wind load

Wind load is calculated as per SBC301, according to parameters as follow:

- Basic wind speed 43 m/s, Exposure class C.
- Topographic factor K_{zt} = 1.0, for no topographic effect is considered.
- Wind directionality factor K_d = 0.95.
- Velocity pressure q_z [km/h] = $0.0473*10^{-3}K_zK_{zt}K_dV^2I$.
- Wind load W [kN] = $q_zGC_fA_f$.

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4.1.5. Seismic load

Seismic load is calculated as per SBC301, according to parameters as follow:

- Site class B.
- 0,2 sec spectral response acceleration S_s = 0.1g, 1,0 sec spectral response acceleration S_1 = 0.02g
- acceleration-based site coefficient (at 0.2-sec period) F_a = 1.0, velocity-based site coefficient (at 1.0-sec period) F_v = 1.0
- Design spectral Response Acceleration at short period S_{DS} = 0.067g, Design spectral Response Acceleration at 1-sec period S_{D1} = 0.131 g

4.2. Load combinations

for allowable stress design:

D+ F_{max}	SBC 301, (Eq. 2.4.1-1)
D+ F_{max}+ W	SBC 301, (Eq. 2.4.1-5)
D+ $F_{operational}$+ 0.7E	SBC 301, (Eq. 2.4.1-5)
0.6D+ W	SBC 301, (Eq. 2.4.1-7)
0.6D+ 0.7E	SBC 301, (Eq. 2.4.1-8)

for strength design:

1.4(D+F)	SBC 301, (Eq. 2.3.2-1)
1.2D+1.6W	SBC 301, (Eq. 2.3.2-1)
1.2D+1.0E	SBC 301, (Eq. 2.3.2-5)
0.9D+1.6W	SBC 301, (Eq. 2.3.2-6)
0.9D+1.0E	SBC 301, (Eq. 2.3.2-7)

5. ALLOWABLE SOIL BEARING CAPACITY

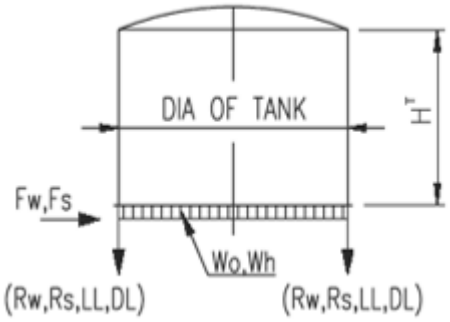
The net safe bearing capacity of soil 375kN/m^2 at 0.6m depth below FGL, as per MD-216-3000-EG-CV-RPT-1008-C01 REPORT 3000, born hole BH-08.

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6. DESIGN METHODOLOGY



Usually, a cylindrical concrete wall is placed to support above industrial liquid tank. The ring wall foundation helps to distribute heavy load and keep stable. The tank will be anchored properly to resist uplifting force caused by wind or earthquake.

Stability of the foundation, and soil pressure under various load conditions are analyzed. Soil capacity and allowable settlement are given based on geotechnical investigation, with factor of safety applied.

7. CALCULATION RESULTS

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7.1. Backwash tank 49W1-TNK-003 foundation design

Loads from tanks are based on drawing number MD-216-49W1-EG-ME-GAD-1005.

diameter of backawsh tank

$D := 6500mm$

empty tank weight per perimeter

$F_{tank} := 127.5kN \quad W_w := \frac{F_{tank}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 6.244 \cdot \frac{kN}{m}$

service platform on tank roof weight per perimeter

$F_{platform} := 40kN \quad W_s := \frac{F_{platform}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 1.959 \cdot \frac{kN}{m}$

operational weight

$F_{ope} := 1726.60kN \quad P_{ope} := \frac{F_{ope}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 52.033 \cdot \frac{kN}{m^2}$

max weight

$F_{max} := 1969.40kN \quad P_{max} := \frac{F_{max}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 59.35 \cdot \frac{kN}{m^2}$

Material property, according to project design criteria MD-216-1000-EG-ST-CRT-1001-A03

concrete specific weight

$\gamma_c := 24 \frac{kN}{m^3}$

concrete compressive strength

$f_{ck} := 35 \frac{N}{mm^2}$

reinforcing bars yield strength

$f_{yk} := 420 \frac{N}{mm^2}$

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Soil data:

soil specfic weight

$$\gamma_s := 20\,\frac{kN}{m^3}$$

Coefficient of active earth pressure

$$K_a := 0.33$$

recommended allowable bearing capacity (BH-08)

$$\sigma_s := 375\,\frac{kN}{m^2}$$

allowable settlement

$$\delta_s := 50mm$$

vertical modulus of subgrade reaction

$$K_s := 33750\,\frac{kN}{m^3}$$

horizontal modulus of subgrade reaction

$$K_h := 155000\,\frac{kN}{m^2}$$

Dimension of ringwall

FGL to bottom of foundation $h_1 := 0.65m$

FGL to top of concrete $h_2 := 0.70m$

depth of ringwall

$$h_R := h_1 + h_2 = 1.35\,m$$

thickness of ringwall $t_R := 400mm$

inner diameter of ring wall

$$D_i := D - t_R = 6.1\,m$$

outer diameter of ring wall

$$D_o := D + t_R = 6.9\cdot m$$

area of ring wall bottom

$$A_R := \frac{\pi\cdot\left(D_o^{\,2} - D_i^{\,2}\right)}{4} = 8.168\,m^2$$

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section modulus for circle

$$S_{circle} := \frac{\pi \cdot \left(D_o^4 - D_i^4\right)}{32 D_o} = 12.551 \cdot m^3$$

Check of soil bearing pressure under the ringwall

Wind load at height of 4600mm above bottom of tank

$$F_{wind} := 39.40kN$$

Overturning moment caused by wind at base

$$M_{wind} := 128.10kN \cdot m$$

stress caused by wind

$$\sigma_{wind} := \frac{M_{wind}}{S_{circle}} = 10.206 \cdot kPa$$

seismic load at base

$$E := 76.70kN$$

Overturning moment caused by earthquake at base

$$M_E := 249.30kN \cdot m$$

stress cause by earthquake

$$\sigma_E := \frac{M_E}{S_{circle}} = 19.863 \cdot kPa$$

soil stress under the ringwall

Empty tank ± wind load

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R + \sigma_{wind} = 63.113 \cdot \frac{kN}{m^2}$$

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R - \sigma_{wind} = 42.7 \cdot \frac{kN}{m^2}$$

Empty tank ± seismic load

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R + 0.7\sigma_E = 66.81 \cdot \frac{kN}{m^2}$$

Full tank

$$\sigma_f := \frac{W_w + W_s}{t_R} + \frac{P_{max}}{2} + \gamma_c \cdot h_R = 82.581 \cdot \frac{kN}{m^2}$$

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Operational load (tank+ operational fluid) ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + \sigma_{wind} = 89.129 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} - \sigma_{wind} = 68.717 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.7\sigma_E = 92.827 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 97.005 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[-\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 81.696 \cdot \frac{kN}{m^2}$$

Full tank ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_{wind} = 92.788 \cdot \frac{kN}{m^2}$$

Full tank + seismic load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_E = 102.444 \cdot \frac{kN}{m^2}$$

Check of overturning at base

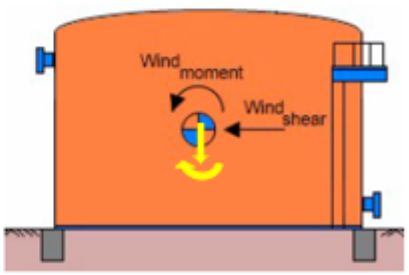
Empty tank ± wind load

Stabilizing moment

$$M_{tank} := F_{tank} \cdot \frac{D}{2} = 414.375 \cdot kN \cdot m$$

$$M_{wind} = 128.1 \cdot kN \cdot m$$

$$\frac{0.9M_{tank}}{1.50M_{wind}} = 1.941$$



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Operational tank ± seismic load

Stabilizing moment

$$M_{ope} := \left(F_{tank} + \pi \cdot D \cdot t_R \cdot h_R \cdot \gamma_c\right) \cdot \frac{D}{2} = 1.274 \times 10^3 \cdot kN \cdot m$$

$$\frac{0.9 M_{ope}}{1.50 M_E} = 3.067$$

Check settlement for ringwall and slab

soil stress under the tank

$$\frac{W_w + W_s}{t_R} + \frac{P_{ope}}{2} + \gamma_c \cdot h_R = 78.923 \cdot \frac{kN}{m^2}$$

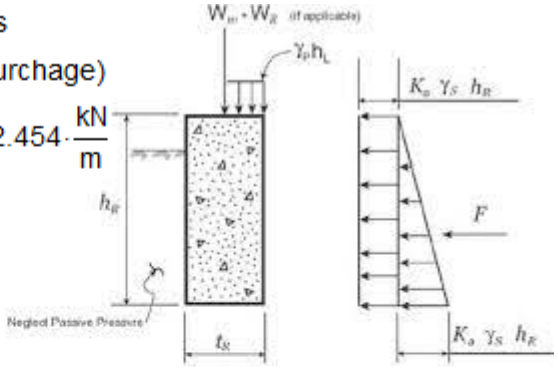
$$P_{ope} + \gamma_s \cdot h_R = 79.033 \cdot \frac{kN}{m^2}$$

soil pressure under ringwall and slab are similar,
no great settlement difference will occur.

Design circumferential reinforcement rings

hoop stress on ringwall (soil pressure + surchage)

$$p := \frac{1}{2} K_a \cdot \gamma_s \cdot h_R^{\,2} + P_{max} \cdot K_a \cdot h_R = 32.454 \cdot \frac{kN}{m}$$



Hoop tension

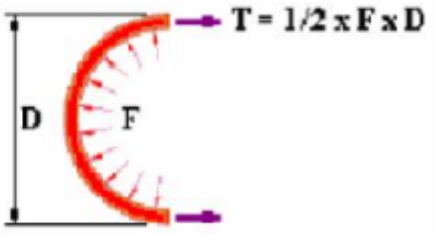
$$T_h := \frac{1}{2} \cdot p \cdot D = 105.477 \cdot kN$$

Deisgn tension

$$T_u := 1.60 T_h = 168.763 \cdot kN$$

Required circumferential reinforcement

$$A_{s.req} := \frac{T_u}{0.87 f_{yk}} = 461.859 \cdot mm^2$$



minimum reinforcement

$$A_{s.min} := 0.0025 \cdot h_R \cdot t_R = 1.35 \times 10^3 \cdot mm^2$$

provide reinforcement ϕ16 mm @150 mm,

$$n := \text{floor}\left(\frac{h_R - 75mm - 40mm}{150mm}\right) = 8$$

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$$A_s := n\pi\cdot\left(\frac{16mm}{2}\right)^2 = 1.608\times10^3\cdot mm^2$$

Design vertical reinforcement

$$A_{v.min} := 0.0018\cdot t_R\cdot\frac{1m}{m} = 720\cdot\frac{mm^2}{m} \quad t_R \text{ in [mm]}$$

provide reinforcement ϕ12 mm @200 mm per face,

$$A_{sp} := 2\left(\frac{1m}{200mm}\right)\pi\cdot\left(\frac{12mm}{2}\right)^2 = 1.131\times10^3\cdot mm^2$$

Moment caused by wind

$M_w := 128.10kN\cdot m$

max. tension force

$$F_T := \frac{M_w}{6.772m} = 18.916\cdot kN$$

Concrete strength in [MPa]

$f_{ck} := 35$

Anchor bolt material F1554 G36

Anchor yield strength

$f_y := 248MPa$

Anchor tensile strength BC304 Appendix D 5.1.2

$f_{uta} := \min(1.9f_y, 860MPa, 400MPa) = 400\cdot MPa$

Anchor bolt diameter

$d := 20mm$



Anchor effective cross section area

$$A_{se} := \pi\cdot\left(\frac{d}{2}\right)^2 = 314.159\cdot mm^2$$

number of anchors

$n := 1$

A02

effective anchor embedment depth in [mm]

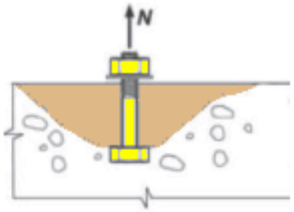
$h_{ef} := 500$

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Tension check SBC304 Appendix D 5.1
nominal strength of anchors in tension



$N_{sa} := n \cdot A_{se} \cdot f_{uta} = 125.664 \cdot kN$

reduction factor for tension SBC304 Appendix D 4.4

$\phi := 0.75$

design strength of tension

$\phi \cdot N_{sa} = 94.248 \cdot kN$

Concrete breakout strength of anchor in tension SBC304 Appendix D 5.2

C01

projected area [mm²] of the failure surface of a single anchor remote from edges:

$A_{NO} := \left(1.5 \cdot h_{ef} + 1.5 \cdot h_{ef}\right) \cdot 400 = 6 \times 10^5$

projected area of the failure surface for the anchor or group of anchors

$A_N := A_{NO}$

factor

$k := 10$

basic concrete breakout strength of a single anchor in tension in [kN]

$$N_b := \frac{k \cdot \sqrt{f_{ck}} \cdot \left(h_{ef}\right)^{1.5}}{1000} = 661.438$$

$$N_{b.max} := \frac{3.8 \cdot \sqrt{f_{ck}} \cdot \left(h_{ef}\right)^{\frac{5}{3}}}{1000} = 708.11$$

modification factor

$\psi_1 := 1.0 \quad c_{min} := 80 \quad \psi_2 := 0.7 + 0.3 \cdot \frac{c_{min}}{1.5 \cdot h_{ef}} = 0.732 \quad \psi_3 := 1.25$

nominal concrete breakout strength in [kN]

$$N_{cbg} := \frac{A_N}{A_{NO}} \cdot \psi_1 \cdot \psi_2 \cdot \psi_3 \cdot N_b = 605.216$$

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7.2. Backwash tank 49W1-TNK-007 foundation design

Loads from tanks are based on drawing number MD-216-49W1-EG-ME-GAD-1006.

diameter of backwash tank

$D := 5000mm$

empty tank weight per perimeter

$F_{tank} := 118kN \quad W_w := \frac{F_{tank}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 7.512 \cdot \frac{kN}{m}$

service platform on tank roof weight per perimeter

$F_{platform} := 24kN \quad W_s := \frac{F_{platform}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 1.528 \cdot \frac{kN}{m}$

operational weight

$F_{ope} := 954kN \quad P_{ope} := \frac{F_{ope}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 48.587 \cdot \frac{kN}{m^2}$

max weight

$F_{max} := 1165.4kN \quad P_{max} := \frac{F_{max}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 59.353 \cdot \frac{kN}{m^2}$

Material property, according to project design criteria MD-216-1000-EG-ST-CRT-1001-A03

concrete specific weight

$\gamma_c := 24 \frac{kN}{m^3}$

concrete compressive strength at 28 days

$f_{ck} := 35 \frac{N}{mm^2}$

reinforcing bars yield strength

$f_{yk} := 420 \frac{N}{mm^2}$

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Soil data:

soil specfic weight

$$\gamma_s := 20\,\frac{kN}{m^3}$$

Coefficient of active earth pressure

$$K_a := 0.33$$

recommended allowable bearing capacity (BH-08)

$$\sigma_s := 375\,\frac{kN}{m^2}$$

allowable settlement

$$\delta_{s.} := 50mm$$

vertical modulus of subgrade reaction

$$K_s := 33750\,\frac{kN}{m^3}$$

horizontal modulus of subgrade reaction

$$K_h := 155000\,\frac{kN}{m^2}$$

Dimension of ringwall

FGL to bottom of foundation $h_1 := 0.65m$

FGL to top of concrete $h_2 := 0.90m$

depth of ringwall

$$h_R := h_1 + h_2 = 1.55\,m$$

thickness of ringwall $t_R := 400mm$

inner diameter of ring wall

$$D_i := D - t_R = 4.6\,m$$

outer diameter of ring wall

$$D_o := D + t_R = 5.4 \cdot m$$

area of ring wall bottom

$$A_R := \frac{\pi \cdot \left(D_o^{\,2} - D_i^{\,2}\right)}{4} = 6.283\,m^2$$

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section modulus for circle

$$S_{circle} := \frac{\pi \cdot \left(D_o^4 - D_i^4\right)}{32 D_o} = 7.319 \cdot m^3$$

Check of soil bearing pressure under the ringwall

Wind load at height of 4300mm above bottom of tank $\frac{5500mm}{2} + h_R = 4.3 \times 10^3 mm$

$$F_{wind} := 26kN$$

Overturning moment caused by wind at base

$$M_{wind} := 84.5kN \cdot m$$

stress caused by wind

$$\sigma_{wind} := \frac{M_{wind}}{S_{circle}} = 11.546 \cdot kPa$$

seismic load at base

$$E := 47kN$$

Overturning moment caused by earthquake at base

$$M_E := 153kN \cdot m$$

stress cause by earthquake

$$\sigma_E := \frac{M_E}{S_{circle}} = 20.905 \cdot kPa$$

soil stress under the ringwall

Empty tank ± wind load

$$\frac{\left(W_w + W_s\right)}{t_R} + \gamma_c \cdot h_R + \sigma_{wind} = 71.346 \cdot \frac{kN}{m^2}$$

$$\frac{\left(W_w + W_s\right)}{t_R} + \gamma_c \cdot h_R - \sigma_{wind} = 48.254 \cdot \frac{kN}{m^2}$$

Empty tank ± seismic load

$$\frac{\left(W_w + W_s\right)}{t_R} + \gamma_c \cdot h_R + 0.7\sigma_E = 74.434 \cdot \frac{kN}{m^2}$$

Full tank

$$\sigma_f := \frac{W_w + W_s}{t_R} + \frac{P_{max}}{2} + \gamma_c \cdot h_R = 89.477 \cdot \frac{kN}{m^2}$$

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Operational load (tank+ operational fluid) ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + \sigma_{wind} = 95.639 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} - \sigma_{wind} = 72.548 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.7\sigma_E = 98.727 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 103.728 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[-\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 86.409 \cdot \frac{kN}{m^2}$$

Full tank ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_{wind} = 101.022 \cdot \frac{kN}{m^2}$$

Full tank + seismicload,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_E = 110.382 \cdot \frac{kN}{m^2}$$

Check of overturning at base

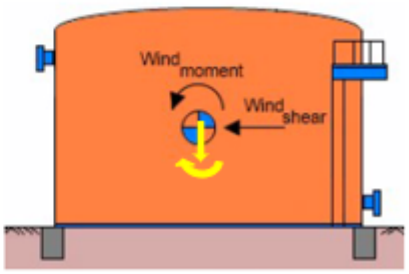
Empty tank ± wind load

Stabilizing moment

$$M_{tank} := F_{tank} \cdot \frac{D}{2} = 295 \cdot kN \cdot m$$

$$M_{wind} = 84.5 \cdot kN \cdot m$$

$$\frac{0.9M_{tank}}{1.50M_{wind}} = 2.095$$



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Operational tank ± seismic load

Stabilizing moment

$$M_{ope} := \left(F_{tank} + \pi \cdot D \cdot t_R \cdot h_R \cdot \gamma_c\right) \cdot \frac{D}{2} = 879.336 \cdot kN \cdot m$$

$$\frac{0.9M_{ope}}{1.50M_E} = 3.448$$

Check settlement for ringwall and slab

soil stress under the tank

$$\frac{W_w + W_s}{t_R} + \frac{P_{ope}}{2} + \gamma_c \cdot h_R = 84.093 \cdot \frac{kN}{m^2}$$

$$P_{ope} + \gamma_s \cdot h_R = 79.587 \cdot \frac{kN}{m^2}$$

soil pressure under ringwall and slab are similar,
no great settlement difference will occur.

Design circumferential reinforcement rings

hoop stress on ringwall (soil pressure + surchage)

$$p := \frac{1}{2} K_a \cdot \gamma_s \cdot h_R^{\,2} + P_{max} \cdot K_a \cdot h_R = 38.287 \cdot \frac{kN}{m}$$

Hoop tension

$$T_h := \frac{1}{2} \cdot p \cdot D = 95.719 \cdot kN$$

Deisgn tension

$$T_u := 1.60 T_h = 153.15 \cdot kN$$

Required circumferential reinforcement

$$A_{s.req} := \frac{T_u}{0.87 f_{yk}} = 419.13 \cdot mm^2$$

minimum reinforcement

$$A_{s.min} := 0.0025 \cdot h_R \cdot t_R = 1.55 \times 10^3 \cdot mm^2$$

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provide reinforcement ϕ16 mm @150 mm,

$$n := \text{floor}\left(\frac{h_R - 75mm - 40mm}{150mm}\right) = 9$$

$$A_s := n\pi\cdot\left(\frac{16mm}{2}\right)^2 = 1.81\times 10^3\cdot mm^2$$

Design vertical reinforcement

$$A_{v.min} := 0.0018\cdot t_R\cdot\frac{1m}{m} = 720\cdot\frac{mm^2}{m} \quad t_R \text{ in [mm]}$$

provide reinforcement ϕ12 mm @200 mm per face,

$$A_{sp} := 2\left(\frac{1m}{200mm}\right)\pi\cdot\left(\frac{12mm}{2}\right)^2 = 1.131\times 10^3\cdot mm^2$$

Moment caused by wind

$$M_W := 84.5kN\cdot m$$

max. tension force

$$F_T := \frac{M_W}{5.3m} = 15.943\cdot kN$$

Concrete strength in [MPa]

$$f_{ck} := 35$$

Anchor bolt material F1554 G36

Anchor yield strength

$$f_y := 248MPa$$

Anchor tensile strength BC304 Appendix D 5.1.2

$$f_{uta} := \min\left(1.9f_y, 860MPa, 400MPa\right) = 400\cdot MPa$$

Anchor bolt diameter

$$d := 20mm$$



Anchor effective cross section area

$$A_{se} := \pi\cdot\left(\frac{d}{2}\right)^2 = 314.159\cdot mm^2$$

number of anchors

$$n := 1$$

effective anchor embedment depth in [mm]

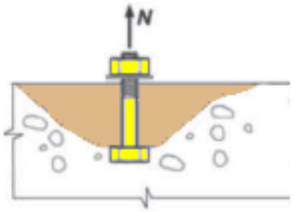
$$h_{ef} := 500$$

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Tension check SBC304 Appendix D 5.1
nominal strength of anchors in tension



$N_{sa} := n \cdot A_{se} \cdot f_{uta} = 125.664 \cdot kN$

reduction factor for tension SBC304 Appendix D 4.4

$\phi := 0.75$

design strength of tension

$\phi \cdot N_{sa} = 94.248 \cdot kN$

Concrete breakout strength of anchor in tension SBC304 Appendix D 5.2

projected area [mm²] of the failure surface of a single anchor remote from edges:

C01

$A_{NO} := (1.5 \cdot h_{ef} + 1.5 \cdot h_{ef}) \cdot 400 = 6 \times 10^5$

projected area of the failure surface for the anchor or group of anchors

$A_N := A_{NO}$

factor

$k := 10$

basic concrete breakout strength of a single anchor in tension in [kN]

$$N_b := \frac{k \cdot \sqrt{f_{ck}} \cdot (h_{ef})^{1.5}}{1000} = 661.438$$

$$N_{b.max} := \frac{3.8 \cdot \sqrt{f_{ck}} \cdot (h_{ef})^{\frac{5}{3}}}{1000} = 708.11$$

modification factor

$\psi_1 := 1.0 \quad c_{min} := 80 \quad \psi_2 := 0.7 + 0.3 \cdot \frac{c_{min}}{1.5 \cdot h_{ef}} = 0.732 \quad \psi_3 := 1.25$

nominal concrete breakout strength in [kN]

$$N_{cbg} := \frac{A_N}{A_{NO}} \cdot \psi_1 \cdot \psi_2 \cdot \psi_3 \cdot N_b = 605.216$$

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7.3. Buffer tank 49W1-REA-003 foundation design

Loads from tanks are based on drawing number MD-216-49W1-EG-ME-GAD-1002 and MD-216-49W1-EG-ME-GAD-1004.

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diameter of buffer tank

$D := 8000mm$

empty tank weight per perimeter

$F_{tank} := 281kN \quad W_w := \frac{F_{tank}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 11.181 \cdot \frac{kN}{m}$

service platform on tank roof weight per perimeter

$F_{platform} := 14kN \quad W_s := \frac{F_{platform}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 0.557 \cdot \frac{kN}{m}$

operational weight

$F_{ope} := 3235kN \quad P_{ope} := \frac{F_{ope}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 64.358 \cdot \frac{kN}{m^2}$

max weight

$F_{max} := 3500kN \quad P_{max} := \frac{F_{max}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 69.63 \cdot \frac{kN}{m^2}$

Material property, according to project design criteria MD-216-1000-EG-ST-CRT-1001-A03

concrete specific weight

$\gamma_c := 24 \frac{kN}{m^3}$

concrete compressive strength at 28 days

$f_{ck} := 35 \frac{N}{mm^2}$

reinforcing bars yield strength

$f_{yk} := 420 \frac{N}{mm^2}$

Soil data:

soil specfic weight

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$$\gamma_s := 20\frac{kN}{m^3}$$

Coefficient of active earth pressure

$$K_a := 0.33$$

recommended allowable bearing capacity (BH-08)

$$\sigma_s := 375\frac{kN}{m^2}$$

allowable settlement

$$\delta_s := 50mm$$

vertical modulus of subgrade reaction

$$K_s := 33750\frac{kN}{m^3}$$

horizontal modulus of subgrade reaction

$$K_h := 155000\frac{kN}{m^2}$$

Dimension of ringwall

FGL to bottom of foundation $h_1 := 0.65m$

FGL to top of concrete $h_2 := 0.75m$

depth of ringwall

$$h_R := h_1 + h_2 = 1.4\,m$$

thickness of ringwall $t_R := 400mm$

inner diameter of ring wall

$$D_i := D - t_R = 7.6\,m$$

outer diameter of ring wall

$$D_o := D + t_R = 8.4\cdot m$$

area of ring wall bottom

$$A_R := \frac{\pi\cdot\left(D_o^{\,2} - D_i^{\,2}\right)}{4} = 10.053\,m^2$$

section modulus for circle

$$S_{circle} := \frac{\pi\cdot\left(D_o^{\,4} - D_i^{\,4}\right)}{32D_o} = 19.197\cdot m^3$$

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Check of soil bearing pressure under the ringwall

Wind load at height of 4600mm above bottom of tank

$F_{wind} := 67.2kN$

Overturning moment caused by wind at base

$M_{wind} := F_{wind} \cdot (5.2m + h_R) = 443.52 \cdot kN \cdot m$

stress caused by wind

$$\sigma_{wind} := \frac{M_{wind}}{S_{circle}} = 23.104 \cdot kPa$$

seismic load at base

$E := 192.6kN$

Overturning moment caused by seismic force
at center of equipment and liquid, seperarely

$$M_E := \frac{F_{tank}}{F_{tank} + F_{ope}} \cdot E \cdot (6850mm + h_R) + \frac{F_{ope}}{F_{tank} + F_{ope}} \cdot E \cdot (4600mm + h_R)$$

$M_E = 1.19 \times 10^3 \cdot kN \cdot m$

stress caused by seismic

$$\sigma_E := \frac{M_E}{S_{circle}} = 62.002 \cdot kPa$$

soil stress under the ringwall
Empty tank ± wind load

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R + \sigma_{wind} = 86.048 \cdot \frac{kN}{m^2}$$

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R - \sigma_{wind} = 39.84 \cdot \frac{kN}{m^2}$$

Empty tank ± seismic load

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R + 0.7\sigma_E = 106.346 \cdot \frac{kN}{m^2}$$

Full tank

$$\sigma_f := \frac{W_w + W_s}{t_R} + \frac{P_{max}}{2} + \gamma_c \cdot h_R = 97.759 \cdot \frac{kN}{m^2}$$

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Operational load (tank+ operational fluid) ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + \sigma_{wind} = 118.227 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} - \sigma_{wind} = 72.019 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.7\sigma_E = 138.525 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 145.003 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[-\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 110.346 \cdot \frac{kN}{m^2}$$

Full tank ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_{wind} = 120.863 \cdot \frac{kN}{m^2}$$

Full tank + seismicload,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_E = 159.762 \cdot \frac{kN}{m^2}$$

Check of overturning at base

Empty tank ± wind load

Stabilizing moment

$$M_{tank} := F_{tank} \cdot \frac{D}{2} = 1.124 \times 10^3 \cdot kN \cdot m$$

$$M_{wind} = 443.52 \cdot kN \cdot m$$

$$\frac{0.9 M_{tank}}{1.50 M_{wind}} = 1.521$$

Operational tank (tank+ operational fluid) ± seismic load

Stabilizing moment

$$M_{ope} := \left(F_{tank} + \pi \cdot D \cdot t_R \cdot h_R \cdot \gamma_c\right) \cdot \frac{D}{2} = 2.475 \times 10^3 \cdot kN \cdot m$$

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$$\frac{0.9M_{ope}}{1.50M_E} = 1.248$$

Check settlement for ringwall and slab

soil stress under the tank

$$\frac{W_w + W_s}{t_R} + \frac{P_{ope}}{2} + \gamma_c \cdot h_R = 95.123 \cdot \frac{kN}{m^2}$$

$$P_{ope} + \gamma_s \cdot h_R = 92.358 \cdot \frac{kN}{m^2}$$

soil pressure under ringwall and slab are similar,
no great settlement difference will occur.

Design circumferential reinforcement rings

hoop stress on ringwall (soil pressure + surchage)

$$p := \frac{1}{2} K_a \cdot \gamma_s \cdot h_R^2 + P_{max} \cdot K_a \cdot h_R = 38.637 \cdot \frac{kN}{m}$$

Hoop tension

$$T_h := \frac{1}{2} \cdot p \cdot D = 154.549 \cdot kN$$

Deisgn tension

$$T_u := 1.60 T_h = 247.278 \cdot kN$$

Required circumferential reinforcement

$$A_{s.req} := \frac{T_u}{0.87 f_{yk}} = 676.732 \cdot mm^2$$

minimum reinforcement

$$A_{s.min} := 0.0025 \cdot h_R \cdot t_R = 1.4 \times 10^3 \cdot mm^2$$

provide reinforcement ϕ16 mm @150 mm,

$$n := \text{floor}\left(\frac{h_R - 75mm - 40mm}{150mm}\right) = 8$$

$$A_{s.p} := n\pi \cdot \left(\frac{16mm}{2}\right)^2 = 1.608 \times 10^3 \cdot mm^2$$

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Design vertical reinforcement. use minimum reinforcement ratio

$$A_{v.min} := 0.0018 \cdot t_R \cdot \frac{1m}{m} = 720 \cdot \frac{mm^2}{m} \quad t_R \text{ in [mm]}$$

provide reinforcement ϕ12 mm @200 mm per face,

$$A_{v.p} := 2\left(\frac{1m}{200mm}\right)\pi \cdot \left(\frac{12mm}{2}\right)^2 = 1.131 \times 10^3 \cdot mm^2$$

Moment caused by wind

$$M_W := 67.2kN \cdot 5.20m = 349.44 \cdot kN \cdot m$$

max. tension force

$$F_T := \frac{M_W}{8.270m} = 42.254 \cdot kN$$

Concrete strength in [MPa]

$$f_{ck} := 35$$

Anchor bolt material F1554 G36

Anchor yield strength

$$f_y := 248MPa$$

Anchor tensile strength BC304 Appendix D 5.1.2

$$f_{uta} := \min\left(1.9f_y, 860MPa, 400MPa\right) = 400 \cdot MPa$$

Anchor bolt diameter

$$d := 20mm$$



Anchor effective cross section area

$$A_{se} := \pi \cdot \left(\frac{d}{2}\right)^2 = 314.159 \cdot mm^2$$

number of anchors

$$n := 1$$

effective anchor embedment depth in [mm]

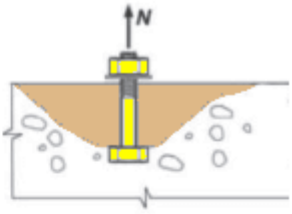
$$h_{ef} := 500$$

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Tension check SBC304 Appendix D 5.1
nominal strength of anchors in tension



$N_{sa} := n \cdot A_{se} \cdot f_{uta} = 125.664 \cdot kN$

reduction factor for tension SBC304 Appendix D 4.4

$\phi := 0.75$

design strength of tension

$\phi \cdot N_{sa} = 94.248 \cdot kN$

Concrete breakout strength of anchor in tension SBC304 Appendix D 5.2

projected area of the failure surface of a single anchor remote from edges:

C01

$A_{NO} := (1.5 \cdot h_{ef} + 1.5 \cdot h_{ef}) \cdot 400 = 6 \times 10^5$

projected area of the failure surface for the anchor or group of anchors

$A_N := A_{NO}$

factor

$k := 10$

basic concrete breakout strength of a single anchor in tension in [kN]

$$N_b := \frac{k \cdot \sqrt{f_{ck}} \cdot (h_{ef})^{1.5}}{1000} = 661.438$$

$$N_{b.max} := \frac{3.8 \cdot \sqrt{f_{ck}} \cdot (h_{ef})^{\frac{5}{3}}}{1000} = 708.11$$

modification factor

$\psi_1 := 1.0 \quad c_{min} := 80 \quad \psi_2 := 0.7 + 0.3 \cdot \frac{c_{min}}{1.5 \cdot h_{ef}} = 0.732 \quad \psi_3 := 1.25$

nominal concrete breakout strength in [kN]

$$N_{cbg} := \frac{A_N}{A_{NO}} \cdot \psi_1 \cdot \psi_2 \cdot \psi_3 \cdot N_b = 605.216$$

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7.4. Buffer tank 49W1-REA-015 foundation design

Loads from tanks are based on drawing number MD-216-49W1-EG-ME-GAD-1001 and MD-216-49W1-EG-ME-GAD-1003.

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diameter of buffer tank (ringwall center-line to center-line)

$D := 6500mm$

empty tank weight per perimeter

$F_{tank} := 252kN \quad W_w := \frac{F_{tank}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 12.341 \cdot \frac{kN}{m}$

service platform on tank roof weight per perimeter

$F_{platform} := 23kN \quad W_s := \frac{F_{platform}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 1.126 \cdot \frac{kN}{m}$

operational weight

$F_{ope} := 3235kN \quad P_{ope} := \frac{F_{ope}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 97.489 \cdot \frac{kN}{m^2}$

max weight

$F_{max} := 3500kN \quad P_{max} := \frac{F_{max}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 105.475 \cdot \frac{kN}{m^2}$

Material property, according to project design criteria
MD-216-1000-EG-ST-CRT-1001-A03

concrete specific weight

$\gamma_c := 24 \frac{kN}{m^3}$

concrete compressive strength at 28 days

$f_{ck} := 35 \frac{N}{mm^2}$

reinforcing bars yield strength

$f_{yk} := 420 \frac{N}{mm^2}$

Soil data:
soil specfic weight

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$\gamma_s := 20\frac{kN}{m^3}$

Coefficient of active earth pressure

$K_a := 0.33$

recommended allowable bearing capacity (BH-08)

$\sigma_s := 375\frac{kN}{m^2}$

allowable settlement

$\delta_s := 50mm$

vertical modulus of subgrade reaction

$K_s := 33750\frac{kN}{m^3}$

horizontal modulus of subgrade reaction

$K_h := 155000\frac{kN}{m^2}$

Dimension of ringwall

FGL to bottom of foundation $h_1 := 0.65m$

FGL to top of concrete $h_2 := 0.75m$

depth of ringwall

$h_R := h_1 + h_2 = 1.4\,m$

thickness of ringwall $t_R := 400mm$

inner diameter of ring wall

$D_i := D - t_R = 6.1\,m$

outer diameter of ring wall

$D_o := D + t_R = 6.9 \cdot m$

area of ring wall bottom

$A_R := \frac{\pi \cdot \left(D_o^2 - D_i^2\right)}{4} = 8.168\,m^2$

section modulus for circle

$S_{circle} := \frac{\pi \cdot \left(D_o^4 - D_i^4\right)}{32 D_o} = 12.551 \cdot m^3$

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Check of soil bearing pressure under the ringwall
Wind load (drawing no. MD-216-49W1-EG-ME-GAD-1003)

$F_{wind} := 62kN$

Overturning moment caused by wind at base

$M_{wind} := F_{wind} \cdot (5.85m + h_R) = 449.5 \cdot kN \cdot m$

stress caused by wind

$$\sigma_{wind} := \frac{M_{wind}}{S_{circle}} = 35.814 \cdot kPa$$

seismic load (drawing no. MD-216-49W1-EG-ME-GAD-1003)

$E := 146.0kN$

Overturning moment caused by seismic force
at center of equipment and liquid, seperarely

$$M_E := \frac{F_{tank}}{F_{tank} + F_{ope}} \cdot E \cdot (8500mm + h_R) + \frac{F_{ope}}{F_{tank} + F_{ope}} \cdot E \cdot (5300mm + h_R)$$

$M_E = 1.012 \times 10^3 \cdot kN \cdot m$

stress caused by seismic

$$\sigma_E := \frac{M_E}{S_{circle}} = 80.627 \cdot kPa$$

soil stress under the ringwall
Empty tank ± wind load

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R + \sigma_{wind} = 103.081 \cdot \frac{kN}{m^2}$$

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R - \sigma_{wind} = 31.454 \cdot \frac{kN}{m^2}$$

Empty tank ± seismic load

$$\frac{(W_w + W_s)}{t_R} + \gamma_c \cdot h_R + 0.7\sigma_E = 123.707 \cdot \frac{kN}{m^2}$$

Full tank

$$\sigma_f := \frac{W_w + W_s}{t_R} + \frac{P_{max}}{2} + \gamma_c \cdot h_R = 120.005 \cdot \frac{kN}{m^2}$$

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Operational load (tank+ operational fluid) ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + \sigma_{wind} = 151.826 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} - \sigma_{wind} = 80.199 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.7\sigma_E = 172.451 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 185.202 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[-\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 131.481 \cdot \frac{kN}{m^2}$$

Full tank ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_{wind} = 155.819 \cdot \frac{kN}{m^2}$$

Full tank + seismicload,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_E = 200.633 \cdot \frac{kN}{m^2}$$

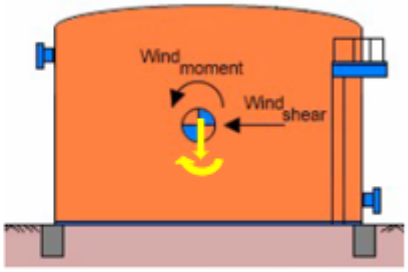
Check of overturning at base

Empty tank ± wind load

Stabilizing moment

$$M_{tank} := F_{tank} \cdot \frac{D}{2} = 819 \cdot kN \cdot m$$

$$\frac{0.9M_{tank}}{1.50M_{wind}} = 1.093$$



Operational tank ± seismic load

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Stabilizing moment

$$M_{ope} := \left(F_{tank} + \pi \cdot D \cdot t_R \cdot h_R \cdot \gamma_c\right) \cdot \frac{D}{2} = 1.711 \times 10^3 \cdot kN \cdot m$$

$$\frac{0.9 M_{ope}}{1.50 M_E} = 1.014$$

Check settlement for ringwall and slab

soil stress under the slab

$$\frac{W_w + W_s}{t_R} + \frac{P_{ope}}{2} + \gamma_c \cdot h_R = 116.012 \cdot \frac{kN}{m^2}$$

$$P_{ope} + \gamma_s \cdot h_R = 125.489 \cdot \frac{kN}{m^2}$$

soil pressure under ringwall and slab are similar,
no great settlement difference will occur.

Design circumferential reinforcement rings

hoop stress on ringwall (soil pressure + surchage)

$$p := \frac{1}{2} K_a \cdot \gamma_s \cdot h_R^2 + P_{max} \cdot K_a \cdot h_R = 55.198 \cdot \frac{kN}{m}$$

Hoop tension

$$T_h := \frac{1}{2} \cdot p \cdot D = 179.392 \cdot kN \qquad T_h = \frac{D}{2} \cdot h_R \cdot K_a \cdot \left[P_{max} + \frac{\left(\gamma_s \cdot h_R\right)}{2}\right]$$

Deisgn tension

$$T_u := 1.60 T_h = 287.028 \cdot kN$$

Required circumferential reinforcement

$$A_{s.req} := \frac{T_u}{0.87 f_{yk}} = 785.517 \cdot mm^2$$

minimum reinforcement

$$A_{s.min} := 0.0025 \cdot h_R \cdot t_R = 1.4 \times 10^3 \cdot mm^2$$

provide reinforcement ϕ16 mm @150 mm,

$$n := \text{floor}\left(\frac{h_R - 75mm - 40mm}{150mm}\right) = 8$$

$$A_s := n\pi \cdot \left(\frac{16mm}{2}\right)^2 = 1.608 \times 10^3 \cdot mm^2$$

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Design vertical reinforcement

max bending moment at bottom

$$M_{bottom} := K_a \cdot P_{max} \cdot \frac{1}{2} \cdot h_R \cdot \left(\frac{1}{2} \cdot h_R \right) + \frac{1}{2} \cdot K_a \cdot \gamma_s \cdot h_R^2 \cdot \frac{1}{3} h_R = 20.074 \cdot kN$$

$$M_u := 1.40 M_{bottom}$$

$$R_n := \frac{M_u \cdot 1m}{0.9 \cdot t_R \cdot h_R^2}$$

$$\rho := \frac{0.85 f_{ck}}{f_{yk}} \cdot \left(1 - \sqrt{1 - \frac{2 \cdot R_n}{0.85 f_{ck}}} \right)$$

$$A_{v.req} := \rho \cdot t_R \cdot h_R = 53.141 \cdot mm^2$$

$$A_{v.min} := 0.0018 \cdot t_R \cdot \frac{1m}{m} = 720 \cdot \frac{mm^2}{m} \quad t_R \text{ in [mm]}$$

provide reinforcement ϕ12 mm @200 mm per face,

$$A_{sp} := 2 \left(\frac{1m}{200mm} \right) \pi \cdot \left(\frac{12mm}{2} \right)^2 = 1.131 \times 10^3 \cdot mm^2$$

Moment caused by wind

$$M_w := 62kN \cdot 5.85m = 362.7 \cdot kN \cdot m$$

max. tension force

$$F_T := \frac{M_w}{6.772m} = 53.559 \cdot kN$$

Concrete strength in [MPa]

$$f_{ck} := 35$$

Anchor bolt material F1554 G36

Anchor yield strength

$$f_y := 248MPa$$

Anchor tensile strength BC304 Appendix D 5.1.2

$$f_{uta} := \min(1.9 f_y, 860MPa, 400MPa) = 400 \cdot MPa$$

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Anchor bolt diameter

$d := 20mm$



Anchor effective cross section area

$$A_{se} := \pi \cdot \left(\frac{d}{2}\right)^2 = 314.159 \cdot mm^2$$

number of anchors

$n := 1$

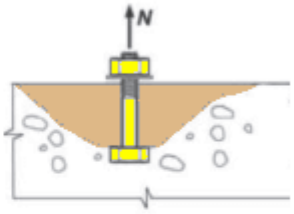
effective anchor embedment depth in [mm]

$h_{ef} := 500$

Tension check SBC304 Appendix D 5.1

nominal strength of anchors in tension

$$N_{sa} := n \cdot A_{se} \cdot f_{uta} = 125.664 \cdot kN$$



reduction factor for tension SBC304 Appendix D 4.4

$\phi := 0.75$

design strength of tension

$$\phi \cdot N_{sa} = 94.248 \cdot kN$$

Concrete breakout strength of anchor in tension SBC304 Appendix D 5.2

projected area of the failure surface of a single anchor remote from edges:

C01

$$A_{NO} := \left(1.5 \cdot h_{ef} + 1.5 \cdot h_{ef}\right) \cdot 400 = 6 \times 10^5$$

projected area of the failure surface for the anchor or group of anchors

$A_N := A_{NO}$

factor

$k := 10$

basic concrete breakout strength of a single anchor in tension in [kN]

$$N_b := \frac{k \cdot \sqrt{f_{ck}} \cdot \left(h_{ef}\right)^{1.5}}{1000} = 661.438$$

$$N_{b.max} := \frac{3.8 \cdot \sqrt{f_{ck}} \cdot \left(h_{ef}\right)^{\frac{5}{3}}}{1000} = 708.11$$

modification factor

$$\psi_1 := 1.0 \quad c_{min} := 80 \quad \psi_2 := 0.7 + 0.3 \cdot \frac{c_{min}}{1.5 \cdot h_{ef}} = 0.732 \quad \psi_3 := 1.25$$

nominal concrete breakout strength in [kN]

$$N_{cbg} := \frac{A_N}{A_{NO}} \cdot \psi_1 \cdot \psi_2 \cdot \psi_3 \cdot N_b = 605.216$$

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7.5. Reject tank 49W1-TNK-009 foundation design

Loads from tanks are based on drawing number MD-216-49W1-EG-ME-GAD-1007.

diameter of buffer tank

$D := 5500mm$

empty tank weight per perimeter

$F_{tank} := 177kN \quad W_w := \frac{F_{tank}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 10.244 \cdot \frac{kN}{m}$

service platform on tank roof weight per perimeter

$F_{platform} := 37kN \quad W_s := \frac{F_{platform}}{\left(2 \cdot \pi \cdot \frac{D}{2}\right)} = 2.141 \cdot \frac{kN}{m}$

operational weight

$F_{ope} := 431.60kN \quad P_{ope} := \frac{F_{ope}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 18.166 \cdot \frac{kN}{m^2}$

max weight

$F_{max} := 809kN \quad P_{max} := \frac{F_{max}}{\pi \cdot \left(\frac{D}{2}\right)^2} = 34.051 \cdot \frac{kN}{m^2}$

Material property, according to project design criteria
MD-216-1000-EG-ST-CRT-1001-A03
concrete specific weight

$\gamma_c := 24 \frac{kN}{m^3}$

concrete compressive strength at 28 days

$f_{ck} := 35 \frac{N}{mm^2}$

reinforcing bars yield strength

$f_{yk} := 420 \frac{N}{mm^2}$

Soil data:
soil specfic weight

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$$\gamma_s := 20\frac{kN}{m^3}$$

Coefficient of active earth pressure

$$K_a := 0.33$$

recommended allowable bearing capacity (BH-08)

$$\sigma_s := 375\frac{kN}{m^2}$$

allowable settlement

$$\delta_s := 50mm$$

vertical modulus of subgrade reaction

$$K_s := 33750\frac{kN}{m^3}$$

horizontal modulus of subgrade reaction

$$K_h := 155000\frac{kN}{m^2}$$

Dimension of ringwall

FGL to bottom of foundation $h_1 := 0.65m$

FGL to top of concrete $h_2 := 0.70m$

depth of ringwall

$$h_R := h_1 + h_2 = 1.35\,m$$

thickness of ringwall $t_R := 400mm$

inner diameter of ring wall

$$D_i := D - t_R = 5.1\,m$$

outer diameter of ring wall

$$D_o := D + t_R = 5.9\cdot m$$

area of ring wall bottom

$$A_R := \frac{\pi\cdot\left(D_o^{\,2} - D_i^{\,2}\right)}{4} = 6.912\,m^2$$

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section modulus for circle

$$S_{circle} := \frac{\pi \cdot \left(D_o^4 - D_i^4\right)}{32 D_o} = 8.906 \cdot m^3$$

Check of soil bearing pressure under the ringwall

Wind load (drawing n0. MD-216-49W1-EG-ME-GAD-1007)

$$F_{wind} := 20kN$$

Overturning moment caused by wind at base

$$M_{wind} := 45.1kN \cdot m$$

stress caused by wind

$$\sigma_{wind} := \frac{M_{wind}}{S_{circle}} = 5.064 \cdot kPa$$

seismic load (drawing n0. MD-216-49W1-EG-ME-GAD-1007)

$$E := 36.5kN$$

Overturning moment caused by seismic force
at center of equipment and liquid, seperarely

$$M_E := 82.1kN \cdot m$$

stress caused by seismic

$$\sigma_E := \frac{M_E}{S_{circle}} = 9.219 \cdot kPa$$

soil stress under the ringwall

Empty tank ± wind load

$$\frac{\left(W_w + W_s\right)}{t_R} + \gamma_c \cdot h_R + \sigma_{wind} = 68.427 \cdot \frac{kN}{m^2}$$

$$\frac{\left(W_w + W_s\right)}{t_R} + \gamma_c \cdot h_R - \sigma_{wind} = 58.299 \cdot \frac{kN}{m^2}$$

Empty tank ± seismic load

$$\frac{\left(W_w + W_s\right)}{t_R} + \gamma_c \cdot h_R + 0.7\sigma_E = 69.816 \cdot \frac{kN}{m^2}$$

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Full tank

$$\frac{W_w + W_s}{t_R} + \frac{P_{max}}{2} + \gamma_c \cdot h_R = 80.389 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + \sigma_{wind} = 77.51 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} - \sigma_{wind} = 67.382 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.7\sigma_E = 78.899 \cdot \frac{kN}{m^2}$$

Operational load (tank+ operational fluid) + seismic load ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 81.084 \cdot \frac{kN}{m^2}$$

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{ope}}{2} + 0.75\left[-\sigma_{wind} + \left(0.7\sigma_E\right)\right] = 73.488 \cdot \frac{kN}{m^2}$$

Full tank ± wind load,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_{wind} = 85.453 \cdot \frac{kN}{m^2}$$

Full tank + seismicload,

$$\frac{W_w + W_s}{t_R} + \gamma_c \cdot h_R + \frac{P_{max}}{2} + \sigma_E = 89.607 \cdot \frac{kN}{m^2}$$

Check of overturning at base

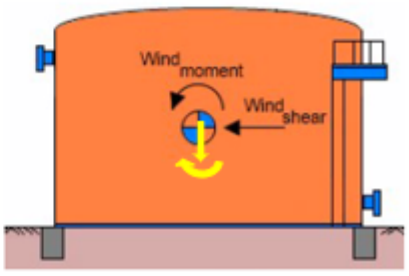
Empty tank ± wind load

Stabilizing moment

$$M_{tank} := F_{tank} \cdot \frac{D}{2} = 486.75 \cdot kN \cdot m$$

$$M_{wind} = 45.1 \cdot kN \cdot m$$

$$\frac{0.9M_{tank}}{1.50M_{wind}} = 6.476$$



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Operational tank (tank+ operational fluid) ± seismic load
Stabilizing moment

$$M_{ope} := \left(F_{tank} + \pi \cdot D \cdot t_R \cdot h_R \cdot \gamma_c\right) \cdot \frac{D}{2} = 1.103 \times 10^3 \cdot kN \cdot m$$

$$\frac{0.9 M_{ope}}{1.50 M_E} = 8.058$$

Check settlement for ringwall and slab
soil stress under the tank

$$\frac{W_w + W_s}{t_R} + \frac{P_{ope}}{2} + \gamma_c \cdot h_R = 72.446 \cdot \frac{kN}{m^2}$$

$$P_{ope} + \gamma_s \cdot h_R = 45.166 \cdot \frac{kN}{m^2}$$

soil pressure under ringwall and slab are similar,
no great settlement difference will occur.

Design circumferential reinforcement rings
hoop stress on ringwall (soil pressure + surchage)

$$p := \frac{1}{2} K_a \cdot \gamma_s \cdot h_R^{\,2} + P_{max} \cdot K_a \cdot h_R = 21.184 \cdot \frac{kN}{m}$$

Hoop tension

$$T_h := \frac{1}{2} \cdot p \cdot D = 58.256 \cdot kN$$

Deisgn tension

$$T_u := 1.60 T_h = 93.21 \cdot kN$$

Required circumferential reinforcement

$$A_{s.req} := \frac{T_u}{0.87 f_{yk}} = 255.09 \cdot mm^2$$

minimum reinforcement

$$A_{s.min} := 0.0025 \cdot h_R \cdot t_R = 1.35 \times 10^3 \cdot mm^2$$

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provide reinforcement ϕ16 mm @150 mm,

$$n := \text{floor}\left(\frac{h_R - 75mm - 40mm}{150mm}\right) = 8$$

$$A_s := n\pi\cdot\left(\frac{16mm}{2}\right)^2 = 1.608\times 10^3\cdot mm^2$$

Design vertical reinforcement

$$A_{v.min} := 0.0018\cdot t_R\cdot\frac{1m}{m} = 720\cdot\frac{mm^2}{m} \quad t_R \text{ in [mm]}$$

provide reinforcement ϕ12 mm @200 mm per face,

$$A_{sp} := 2\left(\frac{1m}{200mm}\right)\pi\cdot\left(\frac{12mm}{2}\right)^2 = 1.131\times 10^3\cdot mm^2$$

Moment caused by wind

$$M_W := 45.1kN\cdot m$$

max. tension force

$$F_T := \frac{M_W}{5.8m} = 7.776\cdot kN$$

Concrete strength in [MPa]

$$f_{ck} := 35$$

Anchor bolt material F1554 G36

Anchor yield strength

$$f_y := 248MPa$$

Anchor tensile strength BC304 Appendix D 5.1.2

$$f_{uta} := \min\left(1.9f_y, 860MPa, 400MPa\right) = 400\cdot MPa$$

Anchor bolt diameter

$$d := 20mm$$



Anchor effective cross section area

$$A_{se} := \pi\cdot\left(\frac{d}{2}\right)^2 = 314.159\cdot mm^2$$

number of anchors

$$n := 1$$

effective anchor embedment depth in [mm]

$$h_{ef} := 500$$

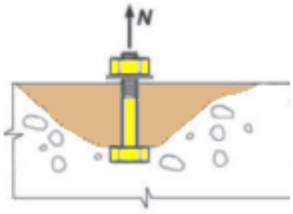
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Tension check SBC304 Appendix D 5.1
nominal strength of anchors in tension

$N_{sa} := n \cdot A_{se} \cdot f_{uta} = 125.664 \cdot kN$



reduction factor for tension SBC304 Appendix D 4.4

$\phi := 0.75$

design strength of tension

$\phi \cdot N_{sa} = 94.248 \cdot kN$

Concrete breakout strength of anchor in tension SBC304 Appendix D 5.2

projected area of the failure surface of a single anchor remote from edges:

C01

$A_{NO} := \left(1.5 \cdot h_{ef} + 1.5 \cdot h_{ef}\right) \cdot 400 = 6 \times 10^5$

projected area of the failure surface for the anchor or group of anchors

$A_N := A_{NO}$

factor

$k := 10$

basic concrete breakout strength of a single anchor in tension in [kN]

$$N_b := \frac{k \cdot \sqrt{f_{ck}} \cdot \left(h_{ef}\right)^{1.5}}{1000} = 661.438$$

$$N_{b.max} := \frac{3.8 \cdot \sqrt{f_{ck}} \cdot \left(h_{ef}\right)^{\frac{5}{3}}}{1000} = 708.11$$

modification factor

$\psi_1 := 1.0 \quad c_{min} := 80 \quad \psi_2 := 0.7 + 0.3 \cdot \frac{c_{min}}{1.5 \cdot h_{ef}} = 0.732 \quad \psi_3 := 1.25$

nominal concrete breakout strength in [kN]

$$N_{cbg} := \frac{A_N}{A_{NO}} \cdot \psi_1 \cdot \psi_2 \cdot \psi_3 \cdot N_b = 605.216$$

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REFERENCES

[1] Saudi Building Code (SBC) 301, Structural - Loading and Forces.

[2] Saudi Building Code (SBC) 304, Structural – Concrete Structures.

[3] Project Design Criteria, MD-216-1000-EG-ST-CRT-1001-A03.

[4] Overall geotechnical report, MD-216-3000-EG-CV-RPT-1008-C01.

[5] 49W1-REA-003 BUFFER TANK 2, MD-216-49W1-EG-ME-GAD-1002

[6] 49W1-REA-003 BUFFER TANK 2, MD-216-49W1-EG-ME-GAD-1004

[7] 49W1-REA-015 BUFFER TANK 1, MD-216-49W1-EG-ME-GAD-1001

[8] 49W1-REA-015 BUFFER TANK 1, MD-216-49W1-EG-ME-GAD-1003

[9] 49W1-TNK-003 BACKWASH TANK, MD-216-49W1-EG-ME-GAD-1005

[10] 49W1-TNK-007 BACKWASH TANK, MD-216-49W1-EG-ME-GAD-1006

[11] 49W1-TNK-007 REJECT TANK, MD-216-49W1-EG-ME-GAD-1007

[12] General arrangement drawing, raw water treatment plan, 1-8/E-J, MD-216-49W1-EG-CV-GAD-1001

[13] General arrangement drawing, raw water treatment plan, 7-12/E-J, MD-216-49W1-EG-CV-GAD-1002

[14] Detail- and reinforcement drawing, raw water treatment plan, buffer tank foundations, MD-216-49W1-EG-CV-DED-1001

[15] Detail- and reinforcement drawing, raw water treatment plan, backwash & reject tank foundations, MD-216-49W1-EG-CV-DED-1002

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ATTACHMENT 1 INPUT DRAWING (FOOTING PLAN AND LOAD DATA)

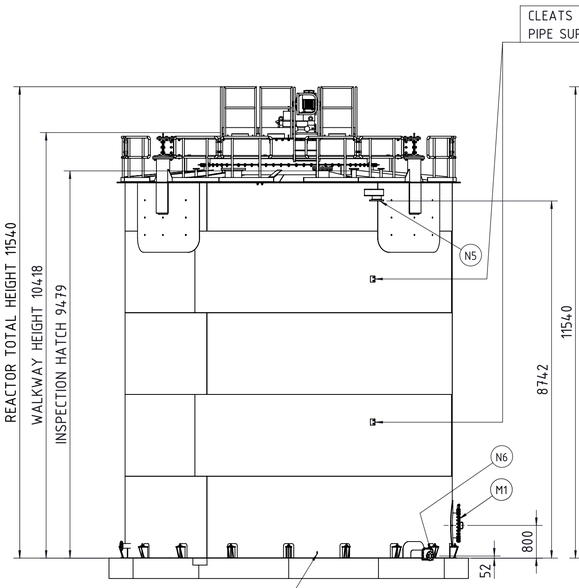
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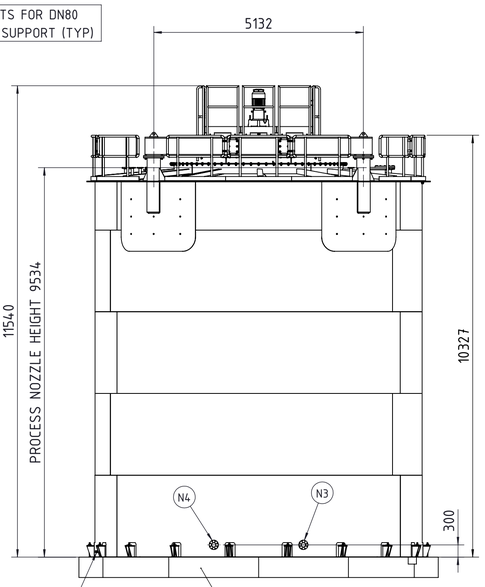
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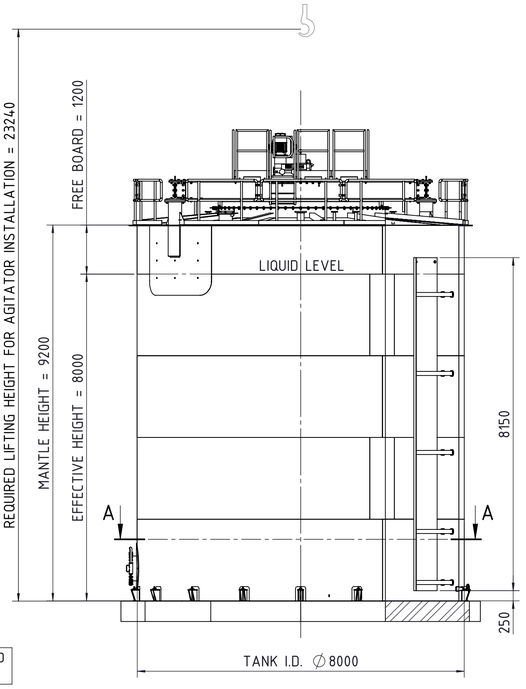
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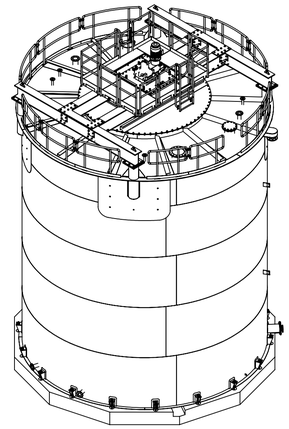
DRAWING NUMBER MD-216-49W1-EG-ME-GAD-1002 | REV A01

NOZZLE TABLE									
No.	SIZE	RATING	DIRECTION	HEIGHT	PROJECTION LENGTH	STANDARD	SERVICE	QUANTITY	REMARKS
N1	16" (DN400)	CLASS 150	115°	9534	3200	ASME B16.5	INLET	1	
N2	2" (DN50)	CLASS 150	45°	9534	3200	ASME B16.5	SUMP INLET	1	
N3	4" (DN100)	CLASS 150	165°	300	4250	ASME B16.5	OUTLET	1	
N4	4" (DN100)	CLASS 150	195°	300	4250	ASME B16.5	OUTLET	1	
N5	3" (DN80)	CLASS 150	60°	8742	4309	ASME B16.5	OVERFLOW	1	FLANGE DOWNWARD
N6	4" (DN100)	CLASS 150	45°	52	4530	ASME B16.5	DRAIN	1	
N7	2" (DN50)	CLASS 150	135°	9534	3200	ASME B16.5	SUMP INLET	1	
N8	2" (DN50)	CLASS 150	245°	9534	3200	ASME B16.5	-	1	
N9	2" (DN50)	CLASS 150	255°	9534	3200	ASME B16.5	NaOH INLET	1	
N13	12" (DN300)	CLASS 150	75°	9534	3200	ASME B16.5	SPARE	1	WITH BLIND FLANGE
N14	4" (DN100)	CLASS 150	60°	9534	3200	ASME B16.5	VENT	1	
K1	6" (DN150)	CLASS 150	270°	9534	3200	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	0°	800	4159	OUTOTEC	MANHOLE	1	
M2	12" (DN300)	-	305°	9534	3200	OUTOTEC	INSPECTION HATCH	1	



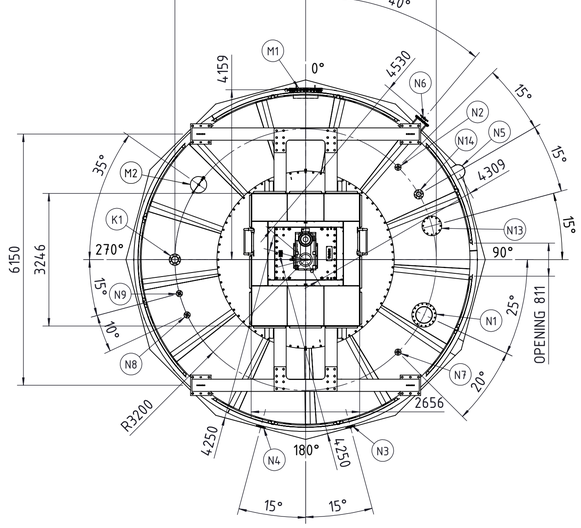


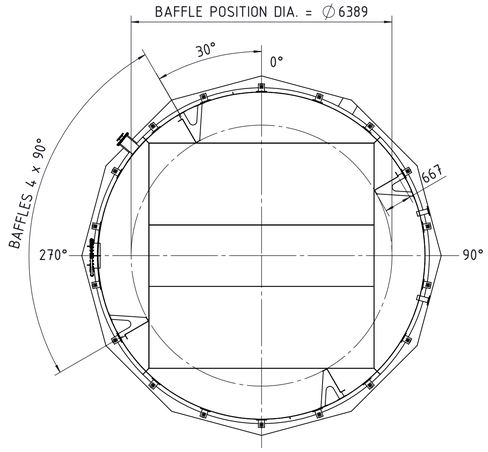




1 : 100

DESIGN DATA	
EQUIPMENT TAG NOs. / NAME PLATE	49W1-REA-003
DESCRIPTION	BUFFER TANK 2
DESIGN CODE	EN 14015 / API 650
DESIGN TEMPERATURE	+5 to +45 °C
DESIGN PRESSURE	+/-8.5mbar
OPERATING PRESSURE	ATM
WIND PRESSURE / CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS (EXCLUDING C.A.)	5 mm
BOTTOM THICKNESS (EXCLUDING C.A.)	5 mm
ROOF THICKNESS (EXCLUDING C.A.)	5 mm
CORROSION ALLOWANCE: SHELL	3 mm
CORROSION ALLOWANCE: BOTTOM	3 mm
CORROSION ALLOWANCE: ROOF	3 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR WITH SANDSTORM
MAIN DIMENSIONS	8400mmx8400mmx11540mm
SHELL HEIGHT	9200 mm
CAPACITY: TOTAL VOLUME	462m3
CAPACITY: LIVE VOLUME	286m3
SEISMIC DESIGN / CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s
SURFACE TREATMENT AND PAINTING	MD-216-3000-EG-ME-SPC-1006
WELDING	AS PER OU500360455





SECTION A-A

PLANT NORTH

MATERIAL OF CONSTRUCTION	
PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A248/ A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS Shore A 40±5

ACID RESISTANT PAINTING TO BE DONE ON INSIDE AND OUTSIDE OF THE TANK:

PAINTING SYSTEM-2, AS PER DOCUMENT MD-216-1000-EG-ME-SPC-1007-A01

NOTE:

1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPL COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN.
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGTH DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGTH DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD, THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.

BUFFER TANK 2 LOAD TABLE	
APP. DEAD WEIGHT (kg)	28100
OPERATIONAL WEIGHT (kg)	433000
MAXIMUM WEIGHT (kg)	493700
AGITATOR UNIT WEIGHT (kg)	10000
SERVICE PLATFORM WEIGHT (kg)	1450
AGITATOR DESIGN TORQUE (Nm)	4707
AGITATOR DESIGN BENDING MOMENT (Nm)	12845

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-49W1-EG-PR-PID-1008	P&I DIAGRAM - BUFFER TANK 2 WITH AGITATOR								
MD-216-1000-EG-ME-SPC-1007-A01	GENERAL SPECIFICATION SURFACE TREATMENT, COATING, PRIMING AND PANTING								
MD-216-49W1-EG-ME-GAD-1004	FOUNDATION AND LOAD PLAN - BUFFER TANK 2 WITH AGITATOR								
MD-216-49W1-EG-PR-LST-1001	EQUIPMENT LIST RAW WATER TREATMENT PLANT (RWTP)								
MD-216-49W1-EG-PR-CRT-1001	PROCESS DESIGN CRITERIA RAW WATER TREATMENT PLANT (RWTP)								
MD-216-1000-EG-CV-DED-1003-C04	DETAIL DRAWING ANCHOR BOLTS STANDARD DETAILS	A01	01.07.2021	ISSUED FOR APPROVAL	AH	TM	JR	GR	

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OUTOTEC DRAWING NUMBER: 900707 GAU01 AG11 MTA01 49002 | DOCUMENT ID: OU601688740 | REVISION: 0

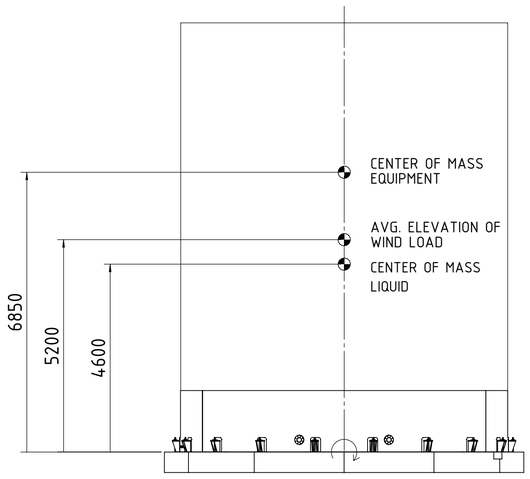
DRAWN	anthyy	01.07.2021
CHECKED	tuoman	01.07.2021
ENGINEER	juhris	01.07.2021
APPROVED	georie	01.07.2021
EMPLOYER		
SCALE	1:75	A1

MANSOURAH-MASSARAH GOLD PROJECT
OUTLINE DRAWING
RAW WATER TREATMENT
49W1-REA-003 BUFFER TANK 2

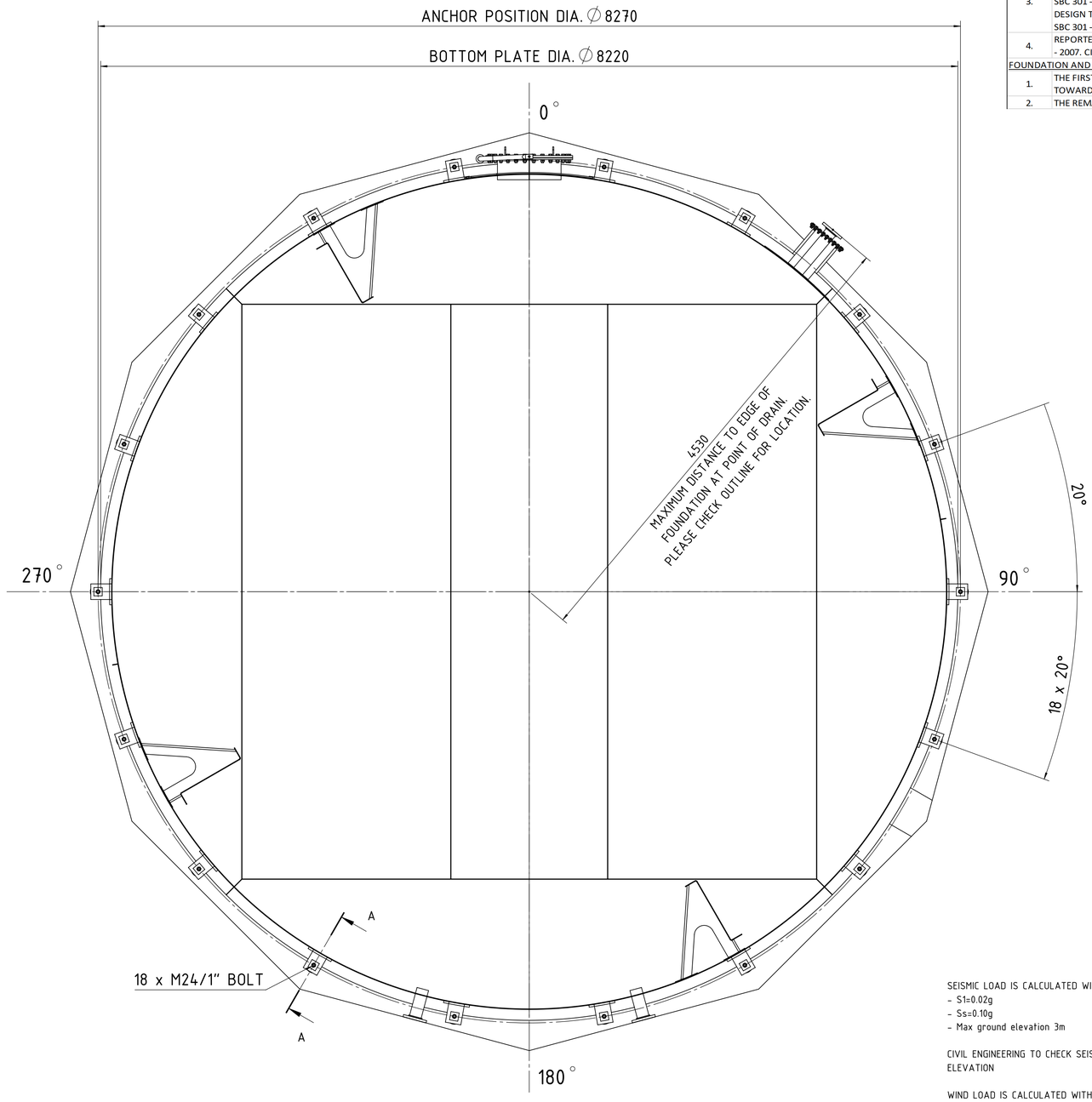
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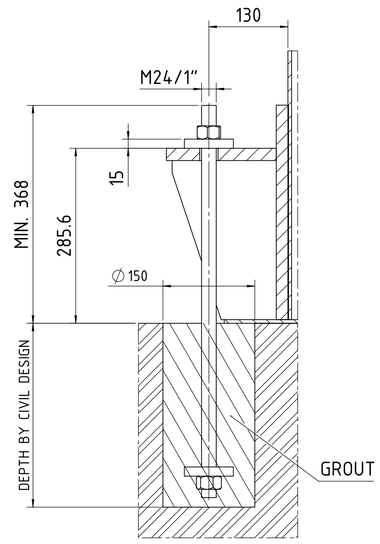
DRAWING NUMBER: 019045_080_M_G_03357_004_R0 | REV: A01

LOADING NOTES:	
1.	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2.	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
3.	REPORTED SEISMIC LOADS ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC 301 - 2007. HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNERACCORDING SBC 301 - 2007.
4.	REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC 301 - 2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL



SCALE 1 : 75





SECTION A-A
SCALE 1 : 5

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES:
- S1=0.02g
- Ss=0.10g
- Max ground elevation 3m

CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

VALUES ARE PRESENTED WITHOUT ANY FACTORS

BUFFER TANK 2	
DEAD WEIGHT	28100 kg
OPERATIONAL WEIGHT OF LIQUID	405000 kg
MAXIMUM WEIGHT OF LIQUID	465700 kg
EXTERNAL PLATFORM LOAD	14 kN
NORMAL OPERATING SPEED	35,1 rpm
WIND LOAD	67,2 kN
SEISMIC LOAD	192,6 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	910,6 kNm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-49W1-EG-PR-PID-1008	P&I DIAGRAM, BUFFER TANK 2 - 49W1-REA-003								
MD-216-49W1-EG-ME-GAD-1002	OUTLINE DRAWING, BUFFER TANK 2 - 49W1-REA-003	A01	17.08.2021	ISSUED FOR APPROVAL	AH	TM	JR	GR	

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OUTOTEC DRAWING NUMBER	900707 GAU01 AG11 MLD15 49002	DOCUMENT ID OU601733635 / REVISION 0

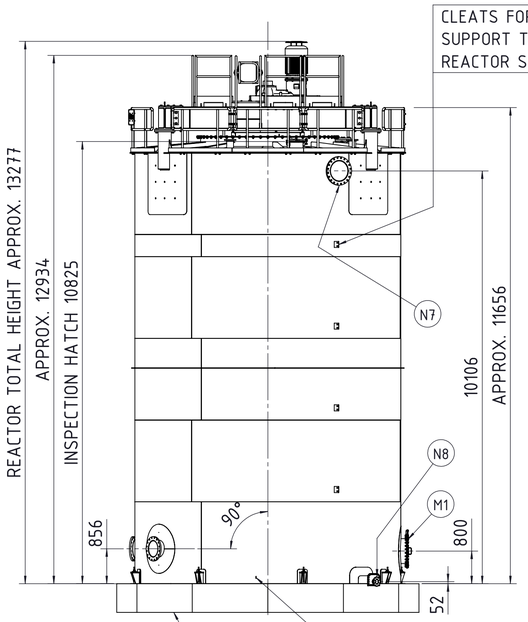
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CHECKED	tuoman	17.08.2021
ENGINEER	juhris	17.08.2021
APPROVED	georie	17.08.2021
EMPLOYER		
SCALE	1:25	A1

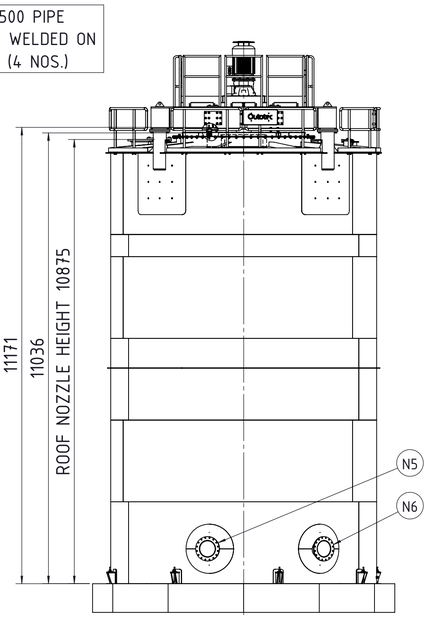
MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
RAW WATER TREATMENT
49W1-REA-003 BUFFER TANK 2

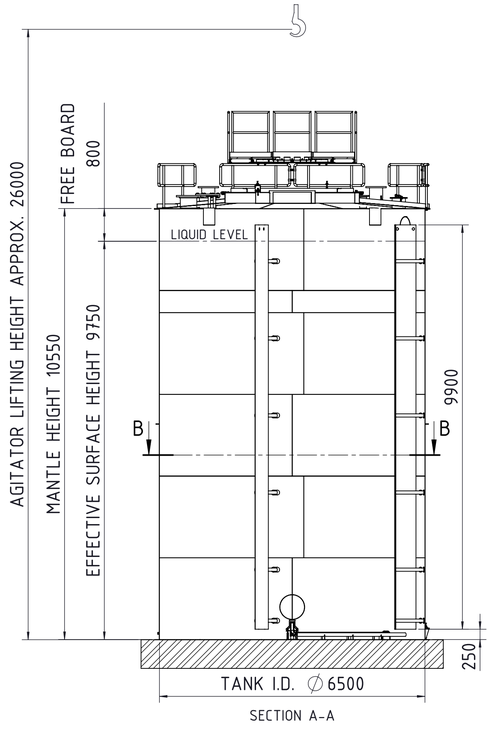
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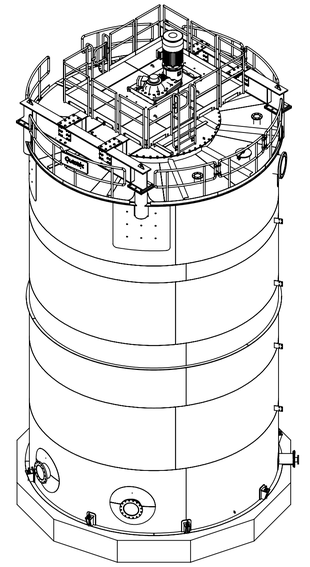
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REV: A01

NOZZLE TABLE									
No.	SIZE	RATING	DIRECTION	HEIGHT	PROJECTION LENGTH	STANDARD	SERVICE	QUANTITY	REMARKS
M1	24" (DN600)	-	0°	800	3412	OUTOTEC	MANHOLE	1	
N1	16" (DN400)	CLASS 150	290°	10875	2700	ASME B16.5	RAW WATER	1	
N2	16" (DN400)	CLASS 150	310°	10875	2700	ASME B16.5	SPARE	1	WITH BLIND FLANGE
N3	2" (DN50)	CLASS 150	285°	10875	2100	ASME B16.5	NaOH INLET	1	
N4	2" (DN50)	CLASS 150	295°	10875	2100	ASME B16.5	RECIRCULATED WATER	1	
N5	16" (DN400)	CLASS 150	195°	856	3430	ASME B16.5	OUTLET	1	
N6	16" (DN400)	CLASS 150	145°	856	3430	ASME B16.5	OUTLET	1	
N7	20" (DN500)	CLASS 150	60°	10106	3500	ASME B16.5	OVERFLOW	1	
N8	4" (DN100)	CLASS 150	45°	52	3780	ASME B16.5	DRAIN	1	
N9	-	-	75°	10825	2700	OUTOTEC	INSPECTION HATCH	1	
N14	6" (DN150)	-	40°	10875	2700	ASME B16.5	AT	1	
K1	6" (DN150)	-	110°	10875	2700	ASME B16.5	LEVEL MEASUREMENT	1	

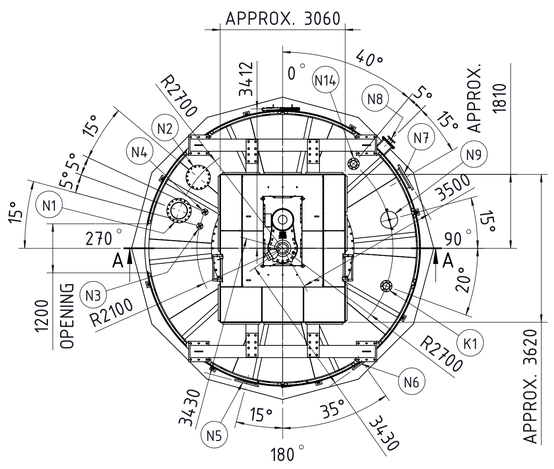


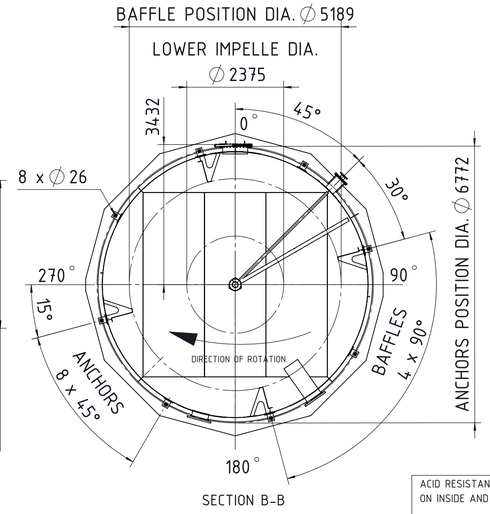






DESIGN DATA	
EQUIPMENT TAG NOs. / NAME PLATE	49W1-REA-015
DESCRIPTION	BUFFER TANK 1
DESIGN CODE	EN 14015 / API 650
DESIGN TEMPERATURE	+5 to +45°C
DESIGN PRESSURE	+/-8.5mbar
OPERATING PRESSURE	ATM
WIND PRESSURE / CODE	Site C / 45 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS (EXCLUDING C.A.)	5 mm
BOTTOM THICKNESS (EXCLUDING C.A.)	5 mm
ROOF THICKNESS (EXCLUDING C.A.)	5 mm
CORROSION ALLOWANCE: SHELL	3 mm
CORROSION ALLOWANCE: BOTTOM	3 mm
CORROSION ALLOWANCE: ROOF	3 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR WITH SANDSTORM
MAIN DIMENSIONS	7400mmx7400mmx13277mm
SHELL HEIGHT	10550 mm
CAPACITY: TOTAL VOLUME	350 m3
CAPACITY: LIVE VOLUME	242 m3
SEISMIC DESIGN / CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s
SURFACE TREATMENT AND PAINTING	MD-216-3000-EG-ME-SPC-1006
WELDING	AS PER OU500360455





MATERIAL OF CONSTRUCTION	
PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A248/ A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS Shore A 40±5

ACID RESISTANT PAINTING TO BE DONE ON INSIDE AND OUTSIDE OF THE TANK:

PAINTING SYSTEM-2, AS PER DOCUMENT MD-216-1000-EG-ME-SPC-1007-A01

NOTE:

1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN.
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGTH DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGTH DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD, THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.

BUFFER TANK 1 LOAD TABLE	
APP. DEAD WEIGHT (kg)	25200
OPERATIONAL WEIGHT (kg)	350000
MAXIMUM WEIGHT (kg)	376500
AGITATOR UNIT WEIGHT (kg)	9000
SERVICE PLATFORM WEIGHT (kg)	1200
AGITATOR DESIGN TORQUE (Nm)	4720
AGITATOR DESIGN BENDING MOMENT (Nm)	15812

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-49W1-EG-PR-PID-1001	P&I DIAGRAM - BUFFER TANK 1 WITH AGITATOR								
MD-216-1000-EG-ME-SPC-1007-A01	GENERAL SPECIFICATION SURFACE TREATMENT, COATING, PRIMING AND PAINTING								
MD-216-49W1-EG-ME-GAD-1003	FOUNDATION AND LOAD PLAN - BUFFER TANK 1 WITH AGITATOR								
MD-216-49W1-EG-PR-LST-1001	EQUIPMENT LIST RAW WATER TREATMENT PLANT (RWTP)								
MD-216-49W1-EG-PR-CRT-1001	PROCESS DESIGN CRITERIA RAW WATER TREATMENT PLANT (RWTP)								
MD-216-1000-EG-CV-DED-1003-C04	DETAIL DRAWING ANCHOR BOLTS STANDARD DETAILS	A01	12.08.2021	ISSUED FOR APPROVAL	AH	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O19045-080-M-G-03356-001-R0 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 GAU01 AG11 MTA01 49001 — DOCUMENT ID: OU601719400 — REVISION: 0

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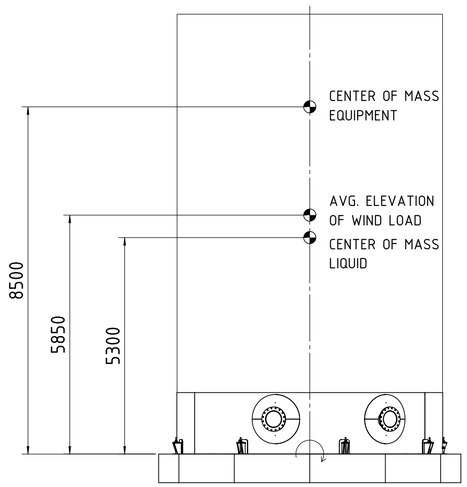
DRAWN	anthyy	17.08.2021
CHECKED	tuoman	17.08.2021
ENGINEER	juhris	17.08.2021
APPROVED	georie	17.08.2021
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
RAW WATER TREATMENT
49W1-REA-015 BUFFER TANK 1

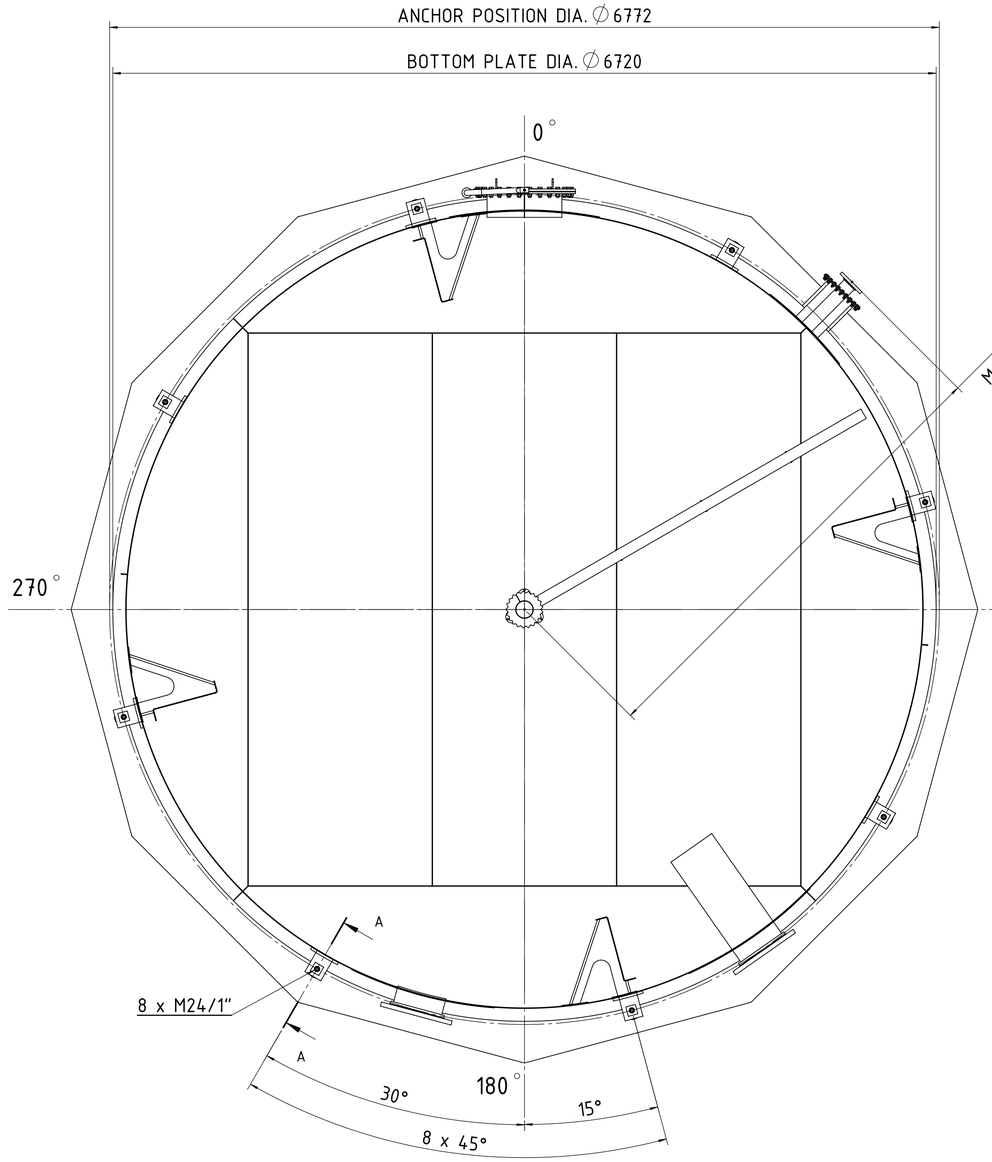
DRAWING NUMBER: MD-216-49W1-EG-ME-GAD-1001 — REV: A01

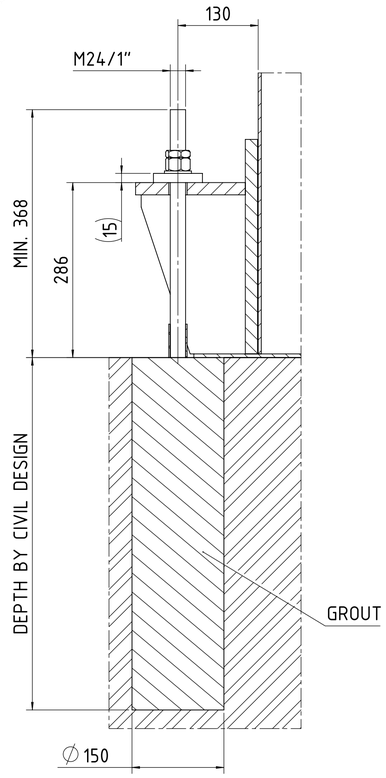
DRAWING NUMBER 019045 080 M G 03357 003 R0 | REV A01

LOADING NOTES:	
1.	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2.	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
3.	REPORTED SEISMIC LOADS ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC 301 - 2007. HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNERACCORDING SBC 301 - 2007.
4.	REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC 301 - 2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL



SCALE 1 : 75





SECTION A-A
SCALE 1 : 5

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES:
- S1=0.02g
- Ss=0.10g
- Max ground elevation 3m

CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

VALUES ARE PRESENTED WITHOUT ANY FACTORS

BUFFER TANK 1	
DEAD WEIGHT	25200 kg
OPERATIONAL WEIGHT OF LIQUID	323500 kg
MAXIMUM WEIGHT OF LIQUID	350000 kg
EXTERNAL PLATFORM LOAD	23 kN
NORMAL OPERATING SPEED	44,5 rpm
WIND LOAD	62 kN
SEISMIC LOAD	146 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	859 kNm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-49W1-EG-PR-PID-1001	P&I DIAGRAM, BUFFER TANK 1 - 49W1-REA-015								
MD-216-49W1-EG-ME-GAD-1001	OUTLINE DRAWING, BUFFER TANK 1 - 49W1-REA-015	A01	16.08.2021	ISSUED FOR APPROVAL	AH	TM	JR	GR	

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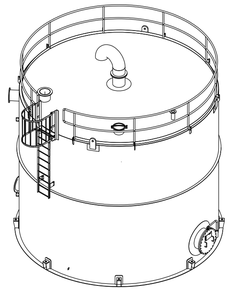
OUTOTEC DRAWING NUMBER: 900707 GAU01 AG11 MLD15 49001 | DOCUMENT ID: 0U601732734 | REVISION: 0

DRAWN	ANTHYY	17.08.2021
CHECKED	tuoman	16.08.2021
ENGINEER	juhris	16.08.2021
APPROVED	georie	16.08.2021
EMPLOYER		
SCALE	1:25	A1

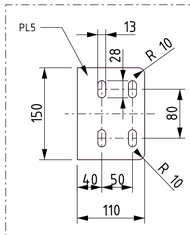
MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
RAW WATER TREATMENT
49W1-REA-015 BUFFER TANK 1

DRAWING NUMBER: MD-216-49W1-EG-ME-GAD-1003 | REV: A01

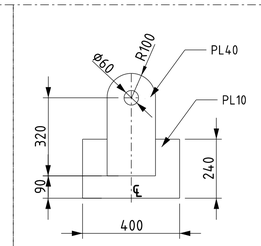
DRAWING NUMBER MD-216-49W1EG-ME-GAD-1005 REV A02



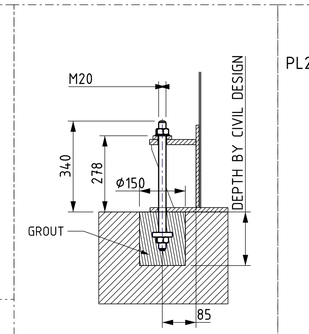
ISOMETRIC VIEW 1:100



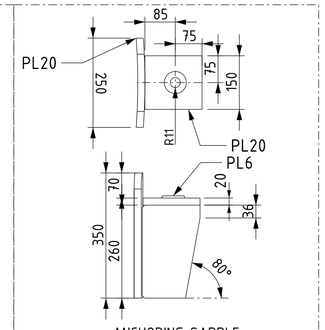
PIPE SUPPORT CLEAT (1:5)



LIFTING LUGS (NOT IN SCALE)



SECTION A-A
NOT IN SCALE



ANCHORING SADDLE
NOT IN SCALE

DESIGN DATA

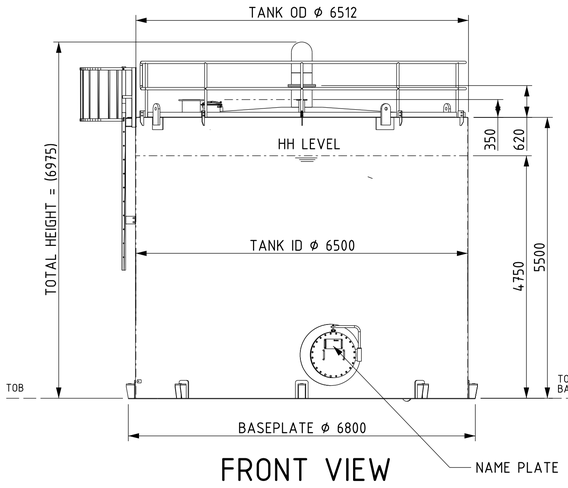
EQUIPMENT TAG NOs. / NAME PLATE	49W1-TNK-003
DESCRIPTION	BACKWASH TANK
DESIGN CODE	API 650 / EN14015
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS (TOTAL)	6 mm
BOTTOM THICKNESS (TOTAL)	8 mm
CORROSION ALLOWANCE: SHELL INSIDE	2 mm
CORROSION ALLOWANCE: BOTTOM	2 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	6975mmx6796mmx7150mm
CAPACITY: FULL VOLUME (A02)	182,5m³
CAPACITY: LIVE VOLUME	160m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s
TANK TYPE	CLOSED ROOF

LOAD DATA

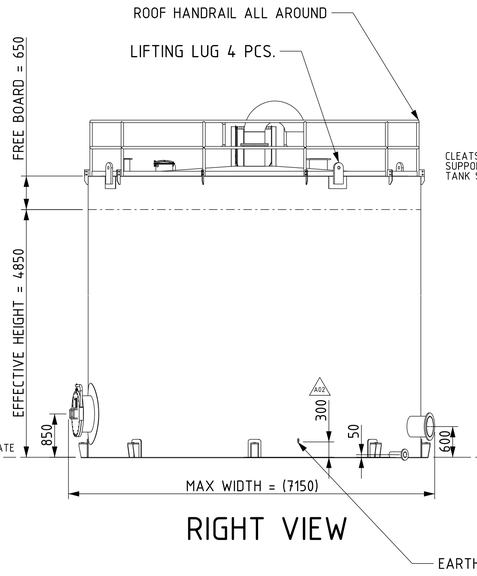
TOTAL WEIGHT (FULL)	195500 kg (A02)
TOTAL WEIGHT (EMPTY)	13000 kg (A02)
FINAL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

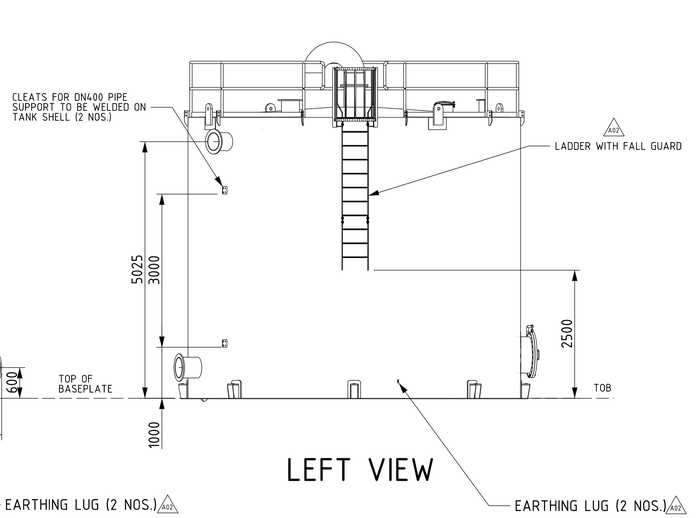
PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR/ S275JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR/ S275/ A36
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
PAINTING	SEE MD-216-1000-EG-ME-SPC-1007



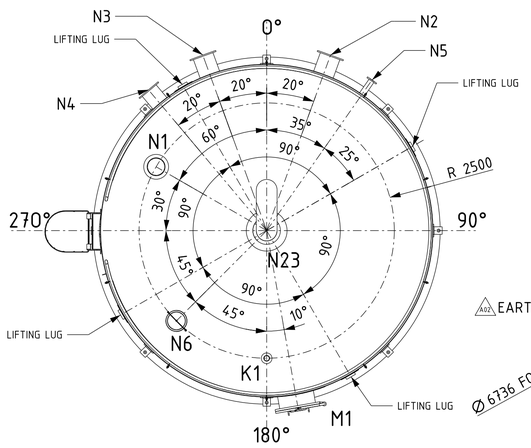
FRONT VIEW



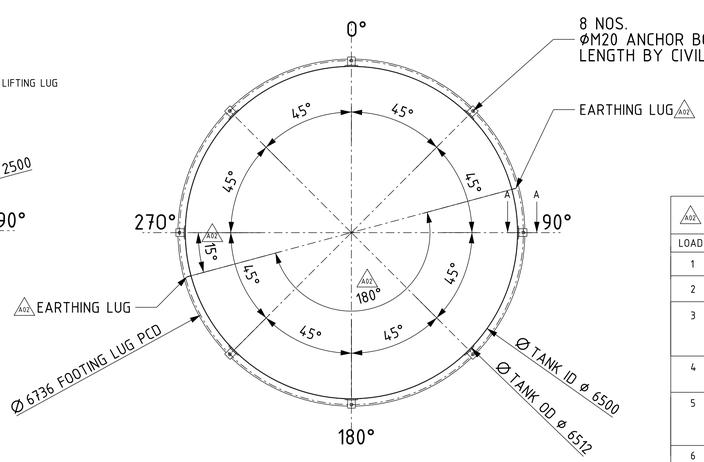
RIGHT VIEW



LEFT VIEW



TOP VIEW



ANCHORING

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
M1	24"		170°	250	3518	850	API 650	MANHOLE
N1	14"	150	300°	350	2500	5850	ASME B16.5	INLET
N2	16"	150	20°	400	3650	600	ASME B16.5	OUTLET
N3	16"	150	340°	400	3650	600	ASME B16.5	OUTLET
N4	16"	150	320°	350	3600	5025	ASME B16.5	OVERFLOW
N5	4"	150	35°	350	3600	50	ASME B16.5	DRAIN
N6	12"		225°	240	2500	5745	API 650	INSPECTION HOLE
N23	16"	150	0°	400	0	6120	ASME B16.5	VENT
K1	4"	150	180°	350	2500	5850	ASME B16.5	LEVEL TRANSMITTER

NOTE:
1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGHT DESIGN
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGHT DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGHT DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
4. ALL STEEL EXTERNAL AND INTERNAL SURFACES SHALL BE BLAST CLEANED TO SA2-5 FINISH. PAINTINGS PER 216-MD-1000-EG-ME-SPC-1007-A01. BOTTOM, SHELL AND ROOF TO BE PAINTED AS PER 216-MD-1000-EG-ME-SPC-1007-A01.
5. NAME PLATE WITH BRACKET AND EARTHING LUGS TO BE PROVIDED BY TANK SUPPLIER.
6. SUFFICIENT TEMPORARY ATTACHEMENT AND SUPPORTS FOR TRANSPORTATION TO BE PROVIDED BY TANK FABRICATOR

FOOTING LOAD TABLE (A02)

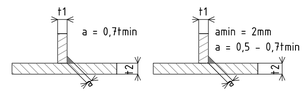
LOAD	DESCRIPTION	VERTICAL (kN)	HORIZONTAL (kN)	MOMENT (kNm)
1	TANK DEAD LOAD	127,50	0,00	0,00
2	NORMAL OPERATING FLUID LOAD	1726,60	0,00	0,00
3	NORMAL OPERATING LOAD (TANK DEAD LOAD + OPERATING FLUID LOAD) (OPERATING CASE)	1 854,10	0,00	0,00
4	WEIGHT OF FULL OF WATER (LOAD)	1969,40	0,00	0,00
5	FULL TANK LOAD (TANK DEAD LOAD + WEIGHT OF FULL OF WATER) (FILL TEST CASE)	2096,90	0,00	0,00
6	SEISMIC LOAD	0,00	76,70	249,3
7	WIND LOAD	0,00	39,40	128,10
8	ROOF LIVE LOAD	40	-	-

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	14-09-21	TANK HEIGHT, LADDER, EARTHING LUGS	TJ	RM	SC	GR	
A01	25-08-21	ISSUED FOR APPROVAL	TJ	RM	SC	GR	

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-ME-GAD-1005	"Anchor bolts, standard details"

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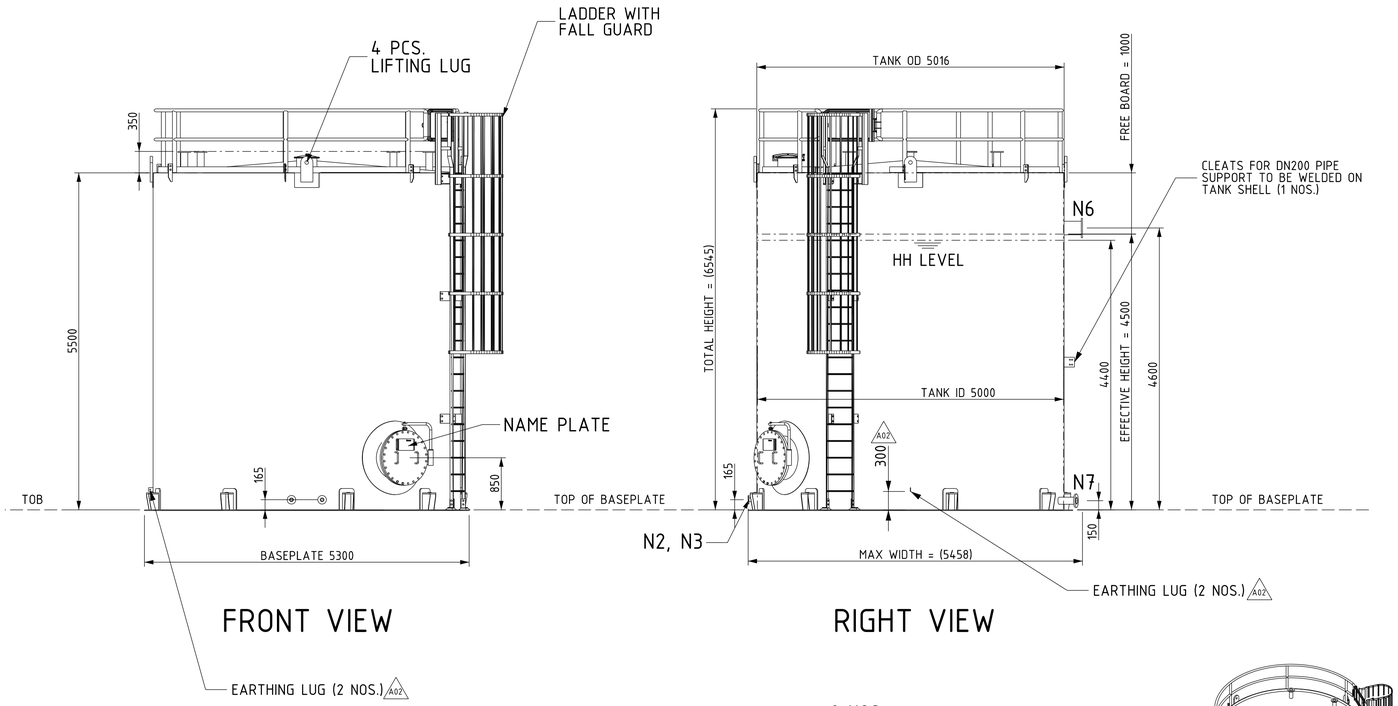
OUTOTEC DRAWING NUMBER: 900707_GAU01_AG11_MTA01_49003 — DOCUMENT ID: OU100728434 — REVISION: 1

DRAWN	TJ	14-09-21
CHECKED	RM	14-09-21
ENGINEER	SC	14-09-21
APPROVED	GR	14-09-21
EMPLOYER		
SCALE	1:60	A1

MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
RAW WATER TREATMENT
49W1-TNK-003 BACKWASH TANK

DRAWING NUMBER: MD-216-49W1-EG-ME-GAD-1005 — REV: A02

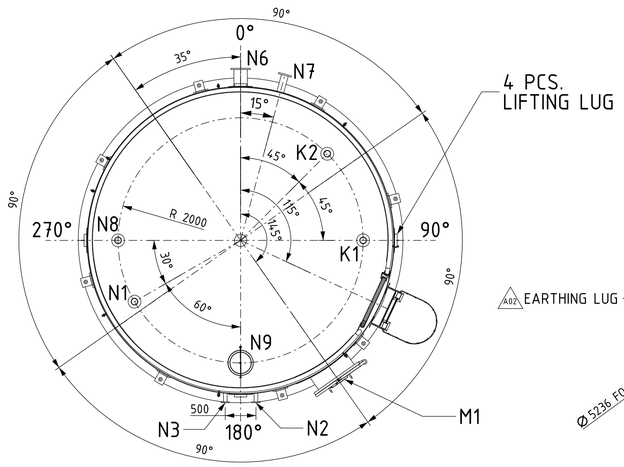
DRAWING NUMBER MD-216-49W1EG-ME-GAD-1006 REV A02



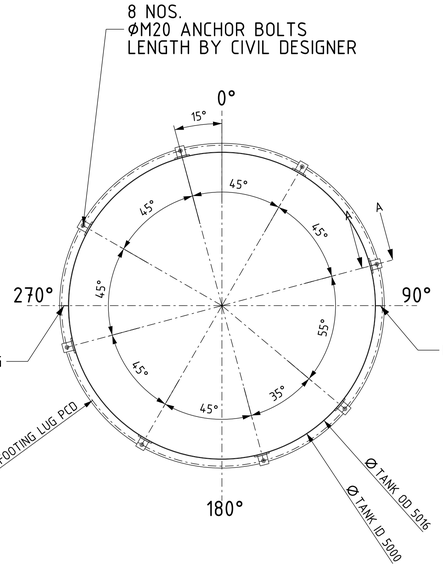
FRONT VIEW

RIGHT VIEW

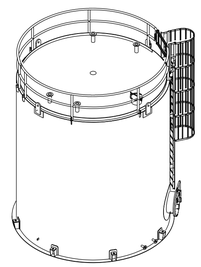
EARTHING LUG (2 NOS.) A02



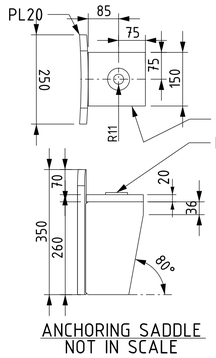
TOP VIEW



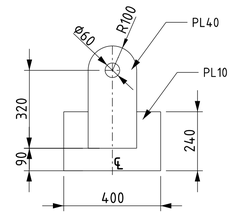
ANCHORING



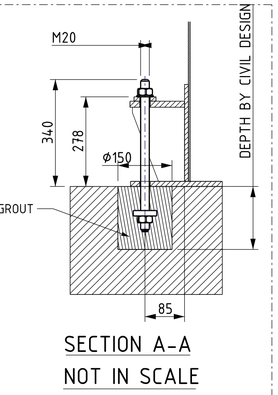
ISOMETRIC VIEW
1:100



ANCHORING SADDLE
NOT IN SCALE



LIFTING LUGS (NOT IN SCALE)



SECTION A-A
NOT IN SCALE

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	49W1-TNK-007
DESCRIPTION	COAGULANT STORAGE TANK
DESIGN CODE	API 650 / EN14015
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS (TOTAL)	8 mm
BOTTOM THICKNESS (TOTAL)	12 mm
CORROSION ALLOWANCE: SHELL INSIDE	2 mm
CORROSION ALLOWANCE: BOTTOM	2 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	6545mmx5458mm
CAPACITY: FULL VOLUME	108m³
CAPACITY: LIVE VOLUME	88.4m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s
TANK TYPE	CLOSED ROOF

LOAD DATA

TOTAL WEIGHT (FULL)	120000 KG
TOTAL WEIGHT (EMPTY)	12000 kg
FINAL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR/ S275JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR/ S275/ A36
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
PAINTING	SEE MD-216-1000-EG-ME-SPC-1007

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
M1	24''		145°	250	2763	850	API 650	MANHOLE
N1	2''	150	240°	350	2000	5850	ASME B16.5	INLET
N2	2''	150	180°	155	2662	165	ASME B16.5	OUTLET
N3	2''	150	180°	155	2662	165	ASME B16.5	OUTLET
N6	8"	150	0°	300	2808	4600	ASME B16.5	OVERFLOW
N7	4''	150	15°	300	2808	150	ASME B16.5	DRAIN
N8	4''	150	270°	350	2000	5850	ASME B16.5	SPARE + BLIND
N9	12''	150	180°	245	2000	5745	API 650	INSPECTION HOLE
K1	4''	150	90°	350	2000	5850	ASME B16.5	LEVEL TRANSMITTER
K2	4''	150	45°	350	2000	5850	ASME B16.5	LEVEL SWITCH

NOTE:
1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGHT DESIGN
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGHT DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGHT DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
4. ALL STEEL EXTERNAL AND INTERNAL SURFACES SHALL BE BLAST CLEANED TO SA2-5 FINISH. PAINTINGS PER 216-MD-1000-EG-ME-SPC-1007-A01. BOTTOM, SHELL AND ROOF TO BE PAINTED AS PER 216-MD-1000-EG-ME-SPC-1007-A01.
5. NAME PLATE WITH BRACKET AND EARTHING LUGS TO BE PROVIDED BY TANK SUPPLIER.
6. SUFFICIENT TEMPORARY ATTACHEMENT AND SUPPORTS FOR TRANSPORTATION TO BE PROVIDED BY TANK FABRICATOR

FOOTING LOAD TABLE

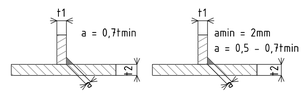
LOAD	DESCRIPTION	VERTICAL (kN)	HORIZONTAL (kN)	MOMENT (kNm)
1	TANK DEAD LOAD	118	0,00	0,00
2	NORMAL OPERATING FLUID LOAD	954	0,00	0,00
3	NORMAL OPERATING LOAD (TANK DEAD LOAD + OPERATING FLUID LOAD) (OPERATING CASE)	1072	0,00	0,00
4	WEIGHT OF FULL OF WATER (LOAD)	1165,4	0,00	0,00
5	FULL TANK LOAD (TANK DEAD LOAD + WEIGHT OF FULL OF WATER) (FILL TEST CASE)	1283,4	0,00	0,00
6	SEISMIC LOAD	0,00	47	153
7	WIND LOAD	0,00	26	84,5
8	ROOF LIVE LOAD	24	-	-

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	01-09-21	CORRECTED AMOUNT OF EARTHING LUGS	TJ	RM	SC	GR	
A01	25-08-21	ISSUED FOR APPROVAL	TJ	RM	SC	GR	

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-ME-GAD-1006	"Anchor bolts, standard details"

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O19045_080_M_G_23515_005_R0 — REVISION: 1

OUTOTEC DRAWING NUMBER: 900707_GAU01_AG11_MTA01_49003 — DOCUMENT ID: OU100728434 — REVISION: 1

DRAWN	TJ	01-09-21
CHECKED	RM	01-09-21
ENGINEER	SC	01-09-21
APPROVED	GR	01-09-21
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
RAW WATER TREATMENT
49W1-TNK-007 COAGULANT STORAGE TANK

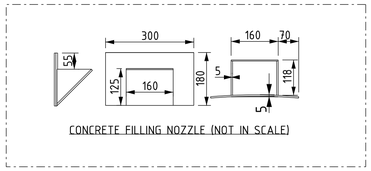
DRAWING NUMBER: MD-216-49W1-EG-ME-GAD-1006 — REV: A02

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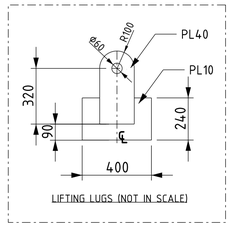
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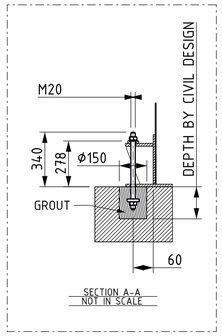
DRAWING NUMBER MD-216-49W1EG-ME-GAD-1007 REV A02



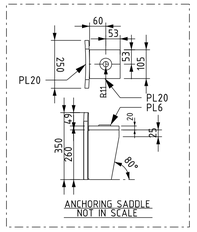
CONCRETE FILLING NOZZLE (NOT IN SCALE)



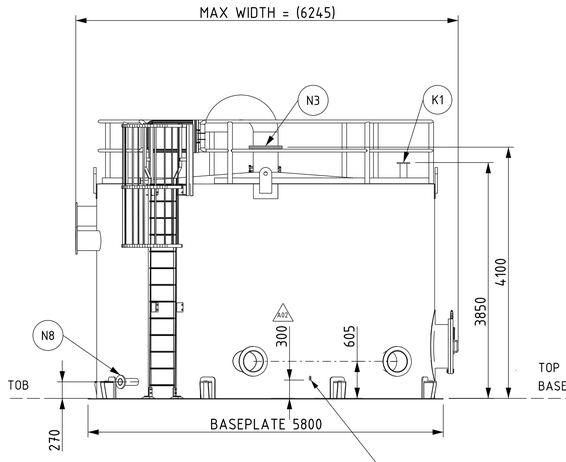
LIFTING LUGS (NOT IN SCALE)



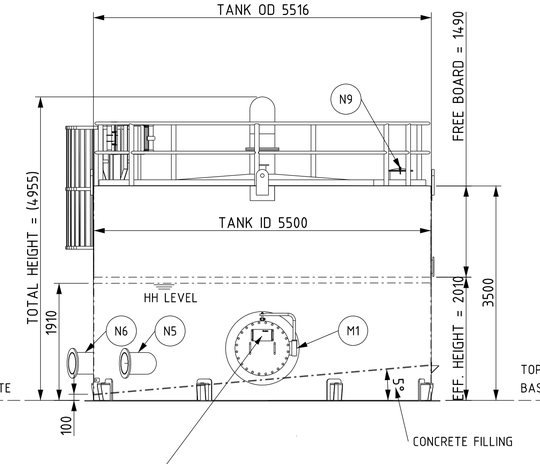
SECTION A-A NOT IN SCALE



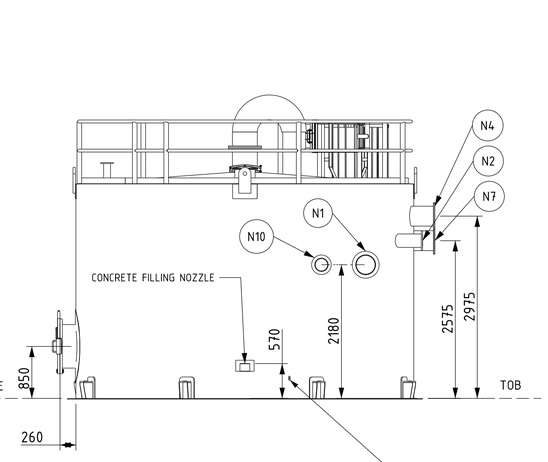
ANCHORING SADDLE NOT IN SCALE



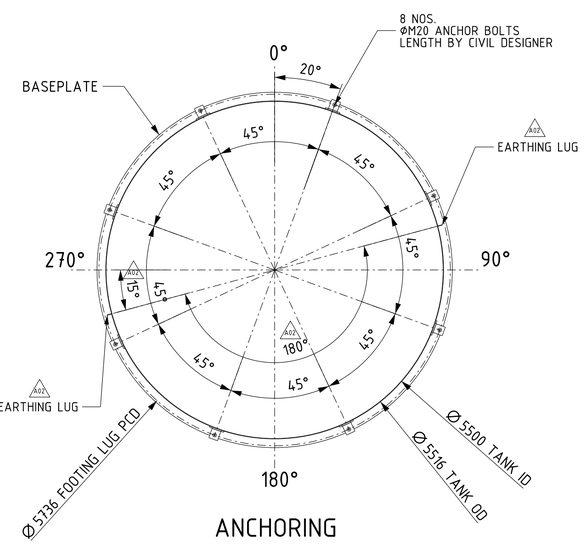
LEFT VIEW



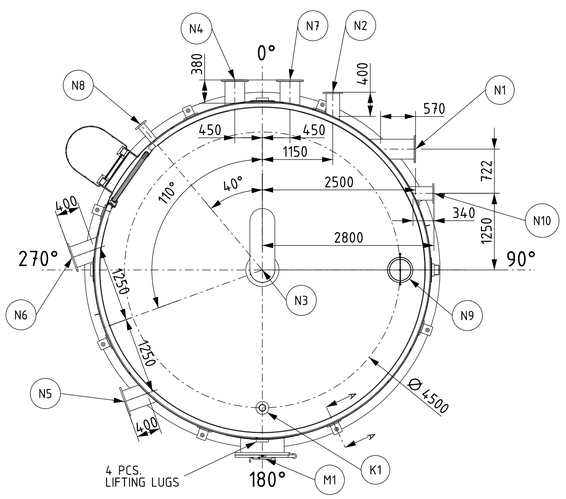
FRONT VIEW



RIGHT VIEW



ANCHORING



TOP VIEW


ISOMETRIC VIEW 1:100

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	49W1-TNK-009
DESCRIPTION	REJECT TANK
DESIGN CODE	API 650 / EN14015
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS (TOTAL)	8 mm
BOTTOM THICKNESS (TOTAL)	12 mm
CORROSION ALLOWANCE: SHELL INSIDE	2 mm
CORROSION ALLOWANCE: BOTTOM	2 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
CAPACITY: FULL VOLUME	75m³
CAPACITY: LIVE VOLUME	40m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s
TANK TYPE	CLOSED ROOF

LOAD DATA

TOTAL WEIGHT (FULL)	93000 kg
TOTAL WEIGHT (EMPTY)	18000 kg

DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
PAINTING	SEE MD-216-1000-EG-ME-SPC-1007

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
M1	24''		180°	250	3020	850	API 650	MANHOLE
N1	12''	150	90°	570		2180	ASME B16.5	INLET
N2	8''	150	0°	400		2575	ASME B16.5	INLET
N3	16''	150		600	0	4100	ASME B16.5	VENT
N4	12''	150	0°	380		2575	ASME B16.5	SPARE+BLIND
N5	14''	150	250°	400		605	ASME B16.5	OUTLET
N6	14''	150	250°	400		605	ASME B16.5	OUTLET
N7	12''	150	0°	380		2975	ASME B16.5	OVERFLOW
N8	4''	150	320°	350	6200	270	ASME B16.5	DRAIN
N9	12''		90°	250	4500	3250	API 650	INSPECTION HOLE
N10	8''	150	90°	340		2180	ASME B16.5	SPARE+BLIND
K1	4''	150	180°	350	4500	3850	ASME B16.5	LEVEL TRANSMITTER

NOTE:
1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGHT DESIGN
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGHT DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGHT DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
4. ALL STEEL EXTERNAL AND INTERNAL SURFACES SHALL BE BLAST CLEANED TO SA2-5 FINISH. PAINTINGS PER 216-MD-1000-EG-ME-SPC-1007-A01. BOTTOM, SHELL AND ROOF TO BE PAINTED AS PER 216-MD-1000-EG-ME-SPC-1007-A01.
5. NAME PLATE WITH BRACKET AND EARTHING LUGS TO BE PROVIDED BY TANK SUPPLIER.
6. SUFFICIENT TEMPORARY ATTACHEMENT AND SUPPORTS FOR TRANSPORTATION TO BE PROVIDED BY TANK FABRICATOR

FOOTING LOAD TABLE

LOAD	DESCRIPTION	VERTICAL (kN)	HORIZONTAL (kN)	MOMENT (kNm)
1	TANK DEAD LOAD	177	0,00	0,00
2	NORMAL OPERATING FLUID LOAD	431,60	0,00	0,00
3	NORMAL OPERATING LOAD (TANK DEAD LOAD + OPERATING FLUID LOAD) (OPERATING CASE)	608,60	0,00	0,00
4	WEIGHT OF FULL OF WATER (LOAD)	809	0,00	0,00
5	FULL TANK LOAD (TANK DEAD LOAD + WEIGHT OF FULL OF WATER) (FILL TEST CASE)	986	0,00	0,00
6	SEISMIC LOAD	0,00	36,5	82,1
7	WIND LOAD	0,00	20	45,1
8	ROOF LIVE LOAD	37	-	-

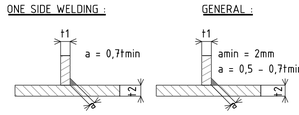
WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : a = 0,7tmin

GENERAL : amin = 2mm, a = 0,5 - 0,7tmin



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	02-09-21	CORRECTED EARTHING LUG AMOUNT	TJ	RM	SC	GR	
MD-216-49W1-EG-ME-GAD-1007	"Anchor bolts, standard details"	A01	27-08-21	ISSUED FOR APPROVAL	TJ	RM	SC	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045 450 M G	1
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707_GAU01_AG11_MTA01_49005	OU100728436 / 1

DRAWN	TJ	02-09-21
CHECKED	RM	02-09-21
ENGINEER	SC	02-09-21
APPROVED	GR	02-09-21
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
RAW WATER TREATMENT
49W1-TNK-009 REJECT TANK

DRAWING NUMBER: MD-216-49W1-EG-ME-GAD-1007 REV: A02

ATTACHMENT 2 COMMENT RESOLUTION SHEET

When comments are received from Employer, they need to be placed in a comment resolution table, and the filled table is always attached to the end of the document.

COMMENT TABLE	Document Name:	Document Number:	Revision No.	Comment Date:
General Comment:				

Sr. No	Chapter	Regarding	Comment	Resolution
1	7.1	project area	(the width of ring wall is only 400 and this will reduce the projected area)	noticed and changed
2	7.2	same as above	same as above	noticed and changed
3	7.3	same as above	same as above	noticed and changed
4	7.4	same as above	same as above	noticed and changed
5	7.5	same as above	same as above	noticed and changed

1 Remember also to always mark the made changes with a triangle to the left side of the document.

You can also reply to the comments on an excel template, then it will be coded as a separate document and submitted to Employer together with the revised document itself.

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You can find the instruction and the excel template from “Document Codes and Numbering Procedure” [1].

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